

BOOK 6

DEFORMATION

6 Deformation of solids

6.1 Stress and strain

Candidates should be able to:

- 1 understand that deformation is caused by tensile or compressive forces (forces and deformations will be assumed to be in one dimension only)
- 2 understand and use the terms load, extension, compression and limit of proportionality
- 3 recall and use Hooke's law
- 4 recall and use the formula for the spring constant $k = F/x$
- 5 define and use the terms stress, strain and the Young modulus
- 6 describe an experiment to determine the Young modulus of a metal in the form of a wire

6.2 Elastic and plastic behaviour

Candidates should be able to:

- 1 understand and use the terms elastic deformation, plastic deformation and elastic limit
- 2 understand that the area under the force–extension graph represents the work done
- 3 determine the elastic potential energy of a material deformed within its limit of proportionality from the area under the force–extension graph
- 4 recall and use $E_p = \frac{1}{2}Fx = \frac{1}{2}kx^2$ for a material deformed within its limit of proportionality

- 1 A spring having spring constant k hangs vertically from a fixed point. A load of weight L , when hung from the spring, causes an extension e . The elastic limit of the spring is not exceeded.

(a) State

- (i) what is meant by an *elastic deformation*,

.....
 [2]

- (ii) the relation between k , L and e .

..... [1]

- (b) Some identical springs, each with spring constant k , are arranged as shown in Fig. 4.1.

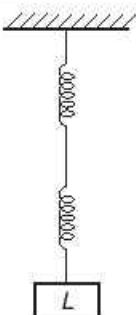
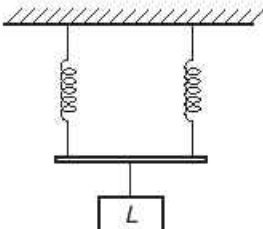
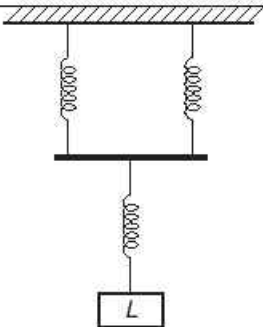
arrangement	total extension	spring constant of arrangement
		
		
		

Fig. 4.1

The load on each of the arrangements is L .

For each arrangement in Fig. 4.1, complete the table by determining

- (i) the total extension in terms of e ,

- (ii) the spring constant in terms of k .

[5]

- 2 A uniform wire has length L and area of cross-section A .

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The wire is fixed at one end so that it hangs vertically with a load attached to its free end, as shown in Fig. 4.1.

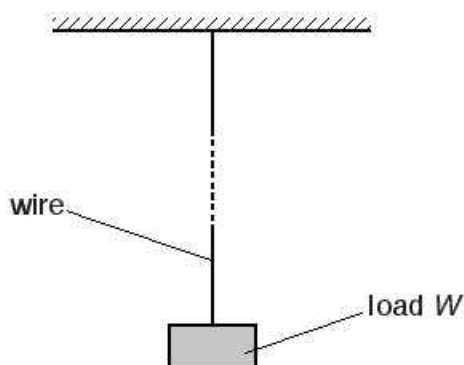


Fig. 4.1

When the load of magnitude W is attached to the wire, it extends by an amount e . The elastic limit of the wire is not exceeded.

The material of the wire has resistivity ρ .

- (a) (i)** Explain what is meant by extends *elastically*.

.....

.....

..... [2]

- (ii)** Write down expressions, in terms of L , A , W , ρ and e for

1. the resistance R of the unstretched wire,

$R = \dots\dots\dots$ [1]

2. the Young modulus E of the wire.

$E = \dots\dots\dots$ [1]

- (b)** A steel wire has resistance $0.44 \, \Omega$. Steel has resistivity $9.2 \times 10^{-8} \, \Omega \text{ m}$.

A load of 34 N hung from the end of the wire causes an extension of $7.7 \times 10^{-4} \text{ m}$.

Using your answers in **(a)(ii)**, calculate the Young modulus E of steel.

$E = \dots\dots\dots \text{ Pa}$ [3]

- 3 (a) Explain what is meant by *strain energy* (*elastic potential energy*).

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.....
.....
..... [2]

- (b) A spring that obeys Hooke's law has a spring constant k .

Show that the energy E stored in the spring when it has been extended elastically by an amount x is given by

$$E = \frac{1}{2}kx^2.$$

[3]

- (c) A light spring of unextended length 14.2 cm is suspended vertically from a fixed point, as illustrated in Fig. 4.1.

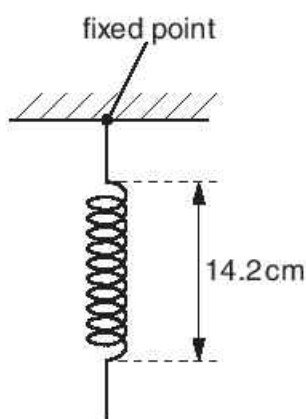


Fig. 4.1

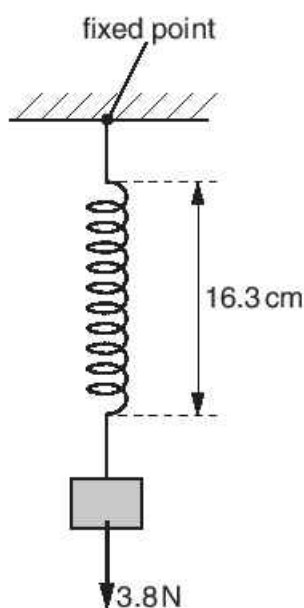


Fig. 4.2

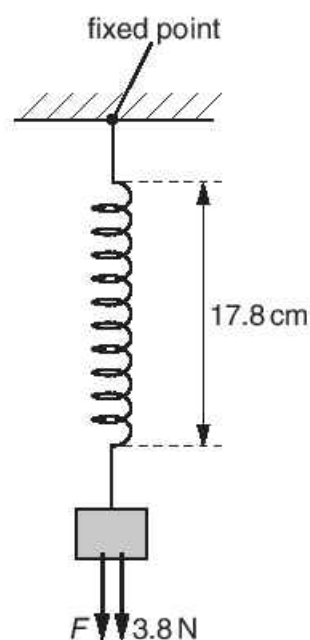


Fig. 4.3

A mass of weight 3.8 N is hung from the end of the spring, as shown in Fig. 4.2. The length of the spring is now 16.3 cm.

An additional force F then extends the spring so that its length becomes 17.8 cm, as shown in Fig. 4.3.

The spring obeys Hooke's law and the elastic limit of the spring is not exceeded.

- (i) Show that the spring constant of the spring is 1.8 N cm^{-1} .

[1]

- (ii) For the extension of the spring from a length of 16.3cm to a length of 17.8cm,
1. calculate the change in the gravitational potential energy of the mass on the spring,

change in energy = J [2]

2. show that the change in elastic potential energy of the spring is 0.077 J,

[1]

3. determine the work done by the force F .

work done = J [1]

- 4 (a) Tensile forces are applied to opposite ends of a copper rod so that the rod is stretched. The variation with stress of the strain of the rod is shown in Fig. 5.1.

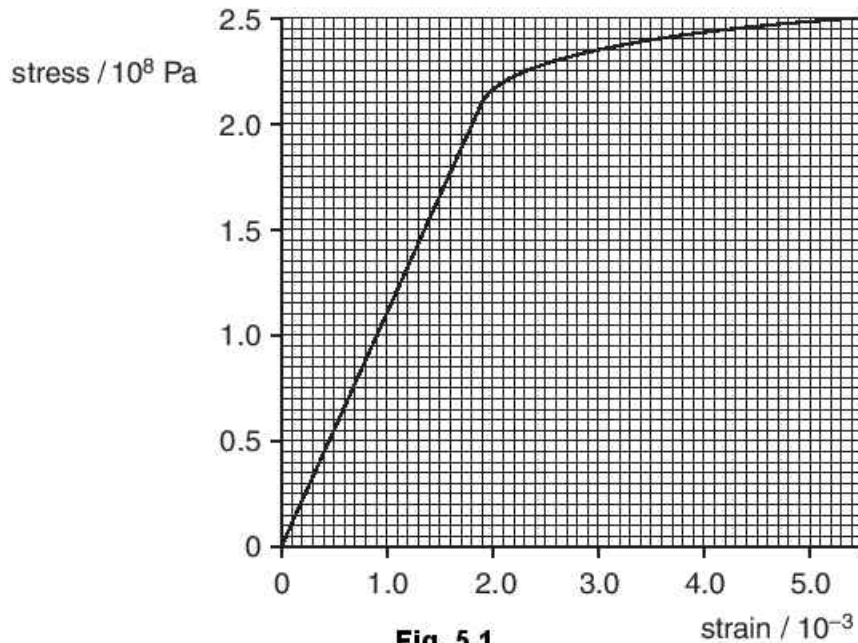


Fig. 5.1

- (i) Use Fig. 5.1 to determine the Young modulus of copper.

Young modulus = Pa [3]

- (ii) On Fig. 5.1, sketch a line to show the variation with stress of the strain of the rod as the stress is reduced from 2.5×10^6 Pa to zero. No further calculations are expected. [1]
- (b) The walls of the tyres on a car are made of a rubber compound. The variation with stress of the strain of a specimen of this rubber compound is shown in Fig. 5.2.

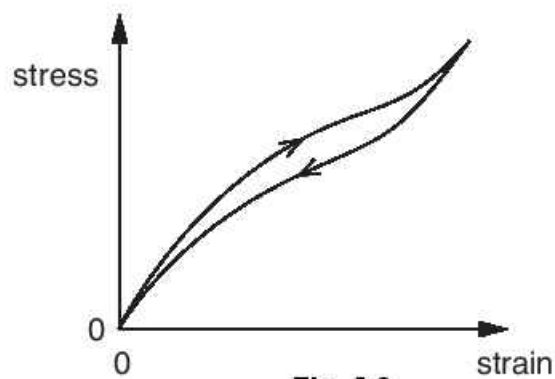


Fig. 5.2

As the car moves, the walls of the tyres bend and straighten continuously. Use Fig. 5.2 to explain why the walls of the tyres become warm.

.....

.....

..... [3]

- 5 (a) A uniform wire has length L and constant area of cross-section A . The material of the wire has Young modulus E and resistivity ρ . A tension F in the wire causes its length to increase by ΔL .

For this wire, state expressions, in terms of L , A , F , ΔL and ρ for

- (i) the stress σ ,

..... [1]

- (ii) the strain ε ,

..... [1]

- (iii) the Young modulus E ,

..... [1]

- (iv) the resistance R .

..... [1]

- (b) One end of a metal wire of length 2.6 m and constant area of cross-section $3.8 \times 10^{-7} \text{ m}^2$ is attached to a fixed point, as shown in Fig. 4.1.

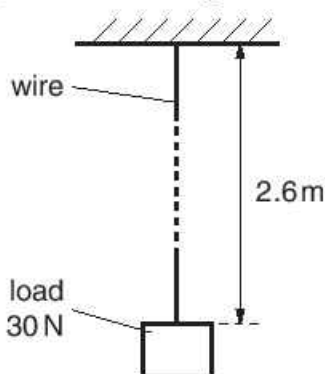


Fig. 4.1

The Young modulus of the material of the wire is $7.0 \times 10^{10} \text{ Pa}$ and its resistivity is $2.6 \times 10^{-8} \Omega \text{ m}$.

A load of 30 N is attached to the lower end of the wire. Assume that the area of cross-section of the wire does not change.

For this load of 30 N,

- (i) show that the extension of the wire is 2.9 mm,

[1]

- (ii) calculate the change in resistance of the wire.

change = Ω [2]

- (c) The resistance of the wire changes with the applied load.

Comment on the suggestion that this change of resistance could be used to measure the magnitude of the load on the wire.

.....

..... [2]

- 6 (a) A metal wire has spring constant k . Forces are applied to the ends of the wire to extend it within the limit of Hooke's law.
Show that, for an extension x , the strain energy E stored in the wire is given by

$$E = \frac{1}{2} kx^2.$$

[4]

- (b) The wire in (a) is now extended beyond its elastic limit. The forces causing the extension are then removed.
The variation with extension x of the tension F in the wire is shown in Fig. 4.1.

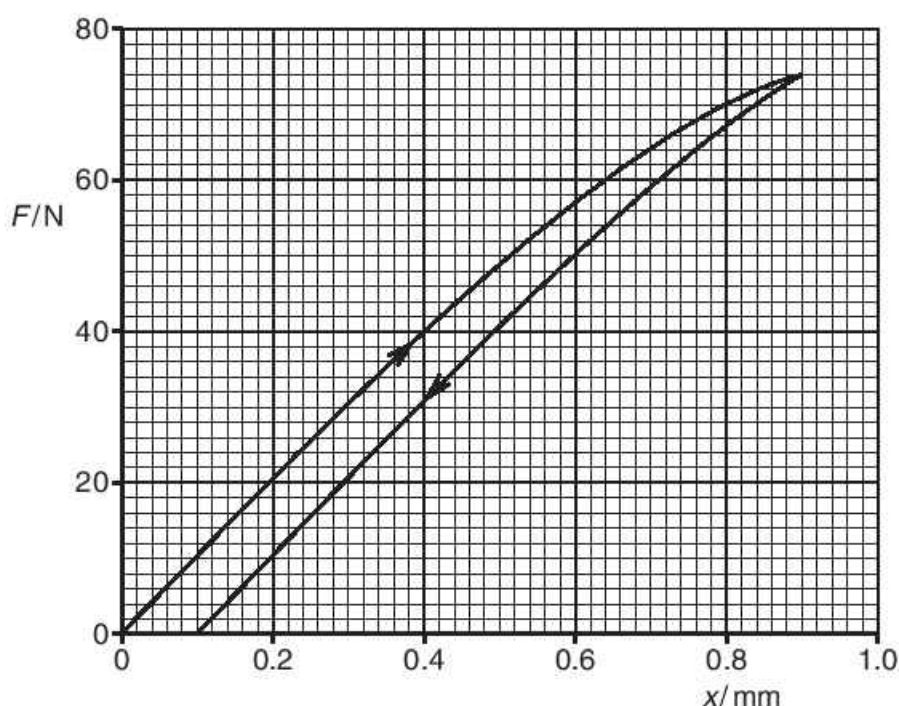


Fig. 4.1

Energy E_S is expended to cause a permanent extension of the wire.

- (i) On Fig. 4.1, shade the area that represents the energy E_S . [1]
- (ii) Use Fig. 4.1 to calculate the energy E_S .

$$E_S = \dots\dots\dots \text{mJ} [3]$$

- (iii) Suggest the change in the structure of the wire that is caused by the energy E_S .

.....
 [1]

7 (a) Define, for a wire,

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(i) *stress*,

.....
..... [1]

(ii) *strain*.

.....
..... [1]

(b) A wire of length 1.70 m hangs vertically from a fixed point, as shown in Fig. 4.1.

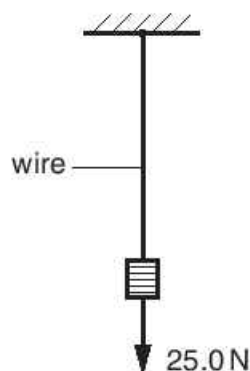


Fig. 4.1

The wire has cross-sectional area $5.74 \times 10^{-8} \text{ m}^2$ and is made of a material that has a Young modulus of $1.60 \times 10^{11} \text{ Pa}$. A load of 25.0 N is hung from the wire.

(i) Calculate the extension of the wire.

extension = m [3]

(ii) The same load is hung from a second wire of the same material. This wire is twice the length but the **same volume** as the first wire. State and explain how the extension of the second wire compares with that of the first wire.

.....
.....
..... [3]

[2]

[4]

- 9 (a) State Hooke's law.

.....
 [1]

- (b) The variation with extension x of the force F for a spring A is shown in Fig. 6.1.

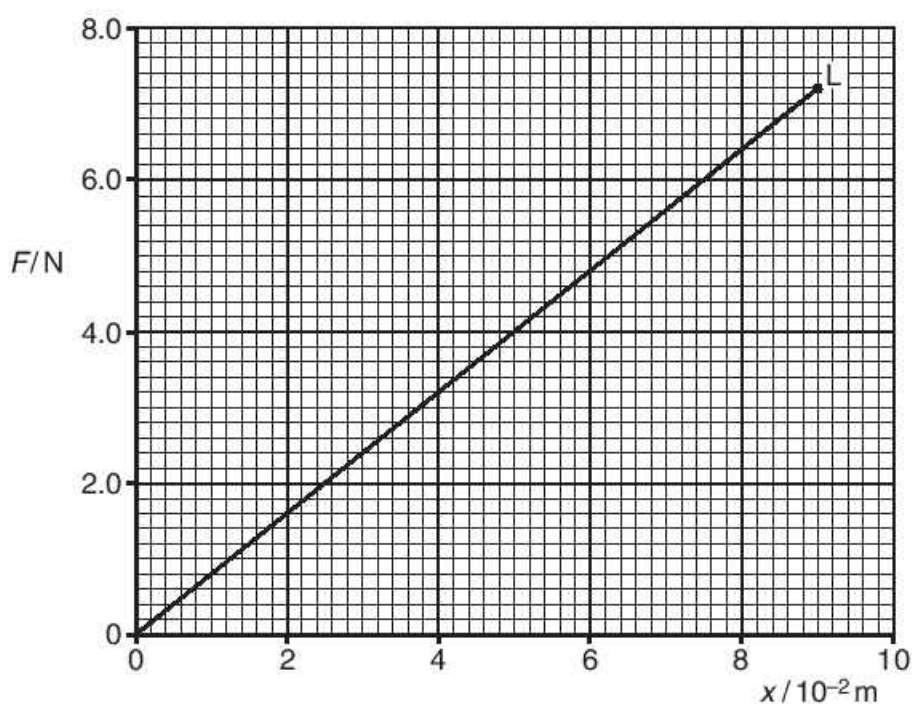


Fig. 6.1

The point L on the graph is the elastic limit of the spring.

- (i) Describe the meaning of *elastic limit*.

.....

 [1]

- (ii) Calculate the spring constant k_A for spring A.

$$k_A = \dots\dots\dots \text{Nm}^{-1} \text{ [1]}$$

- (iii) Calculate the work done in extending the spring with a force of 6.4 N.

work done = J [2]

- (c) A second spring B of spring constant $2k_A$ is now joined to spring A, as shown in Fig. 6.2.

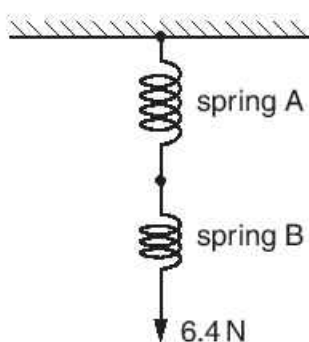


Fig. 6.2

A force of 6.4 N extends the combination of springs.

For the combination of springs, calculate

- (i) the total extension,

extension = m [1]

- (ii) the spring constant.

spring constant = Nm^{-1} [1]

10 (a) Define

9702/22/O/N/11/Q3

(i) *stress*,

.....
.....[1]

(ii) *strain*.

.....
.....[1]

(b) Explain the term *elastic limit*.

.....
.....[1]

(c) Explain the term *ultimate tensile stress*.

.....
.....
.....[2]

(d) (i) A **ductile** material in the form of a wire is stretched up to its breaking point. On Fig. 3.1, sketch the variation with extension x of the stretching force F .



Fig. 3.1

[2]

- (ii) On Fig. 3.2, sketch the variation with x of F for a **brittle** material up to its breaking point.



Fig. 3.2

[1]

- (e) (i) Explain the features of the graphs in (d) that show the characteristics of ductile and brittle materials.

.....

.....

.....

.....

..... [2]

- (ii) The force F is removed from the materials in (d) just before the breaking point is reached. Describe the subsequent change in the extension for

1. the ductile material,

.....

..... [1]

2. the brittle material.

.....

..... [1]

- 11 One end of a spring is fixed to a support. A mass is attached to the other end of the spring. The arrangement is shown in Fig. 3.1.

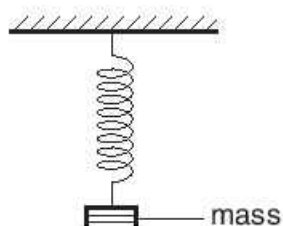


Fig. 3.1

- (a) The mass is in equilibrium. Explain, by reference to the forces acting on the mass, what is meant by equilibrium.

.....

 [2]

- (b) The mass is pulled down and then released at time $t = 0$. The mass oscillates up and down. The variation with t of the displacement of the mass d is shown in Fig. 3.2.

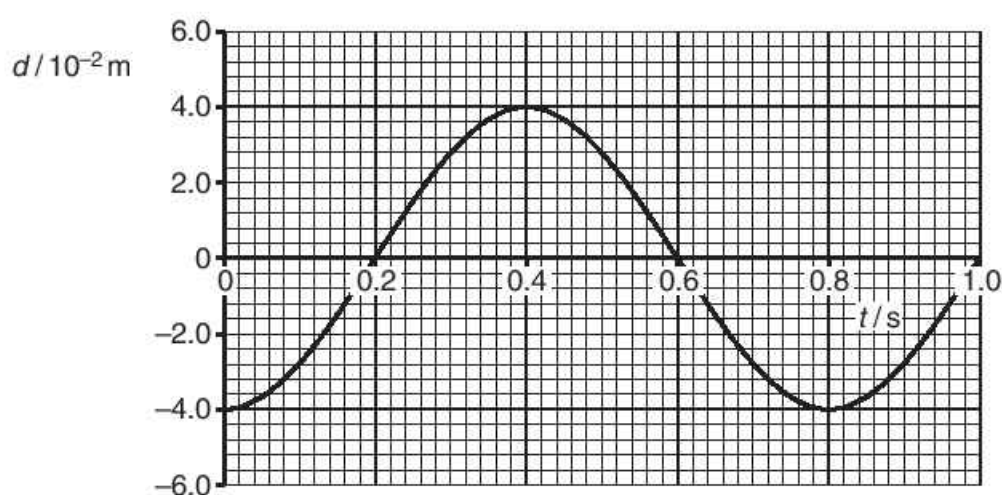


Fig. 3.2

Use Fig. 3.2 to state a time, one in each case, when

- (i) the mass is at maximum speed,

time = s [1]

- (ii) the elastic potential energy stored in the spring is a maximum,

time = s [1]

- (iii) the mass is in equilibrium.

time = s [1]

- (c) The arrangement shown in Fig. 3.3 is used to determine the length l of a spring when different masses M are attached to the spring.

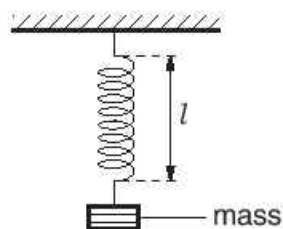


Fig. 3.3

The variation with mass M of l is shown in Fig. 3.4.

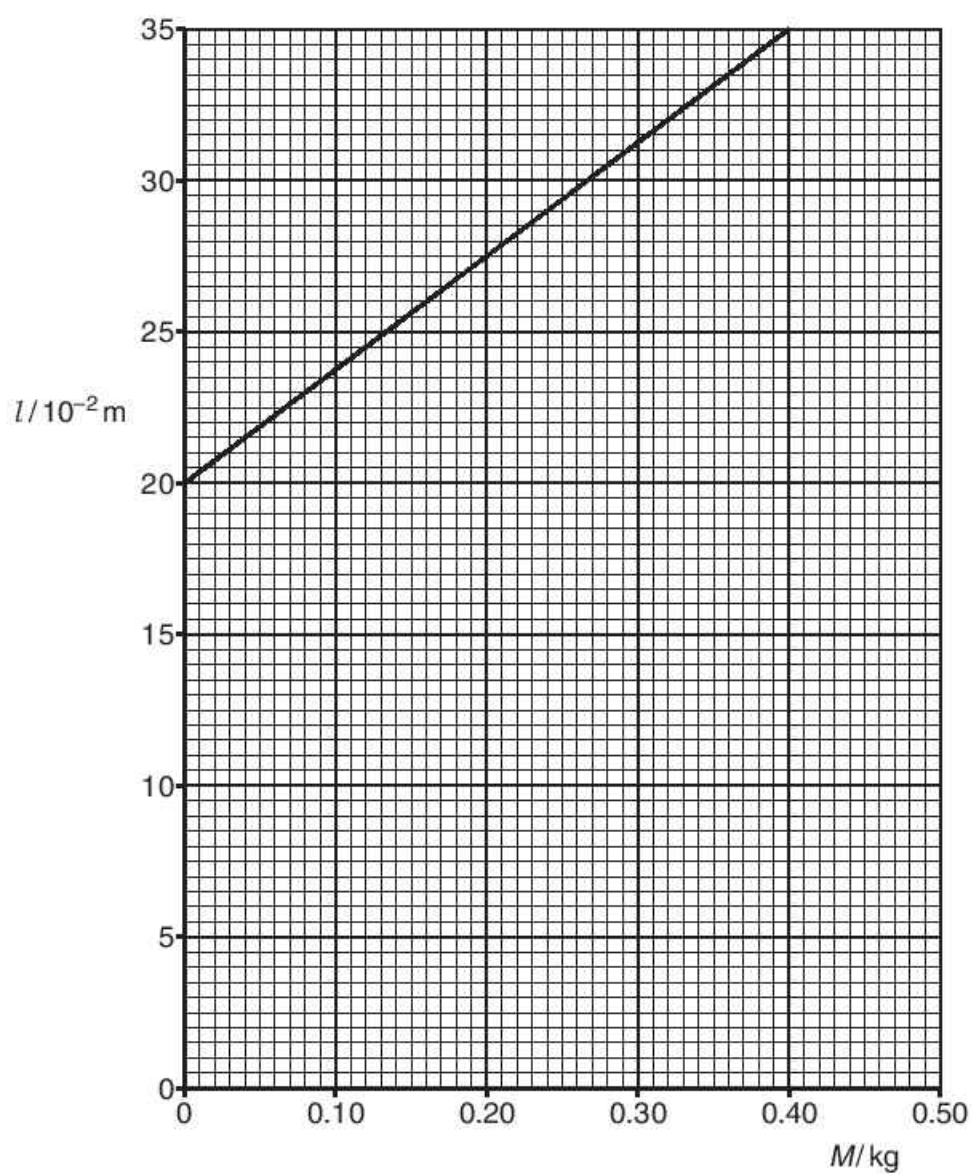


Fig. 3.4

- (i) State and explain whether the spring obeys Hooke's law.

.....
.....
..... [2]

- (ii) Show that the force constant of the spring is 26 N m^{-1} .

[2]

- (iii) A mass of 0.40 kg is attached to the spring. Calculate the energy stored in the spring.

energy = J [3]

12 (a) Define the *Young modulus*.

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.....
 [1]

- (b) A load F is suspended from a fixed point by a steel wire. The variation with extension x of F for the wire is shown in Fig. 5.1.

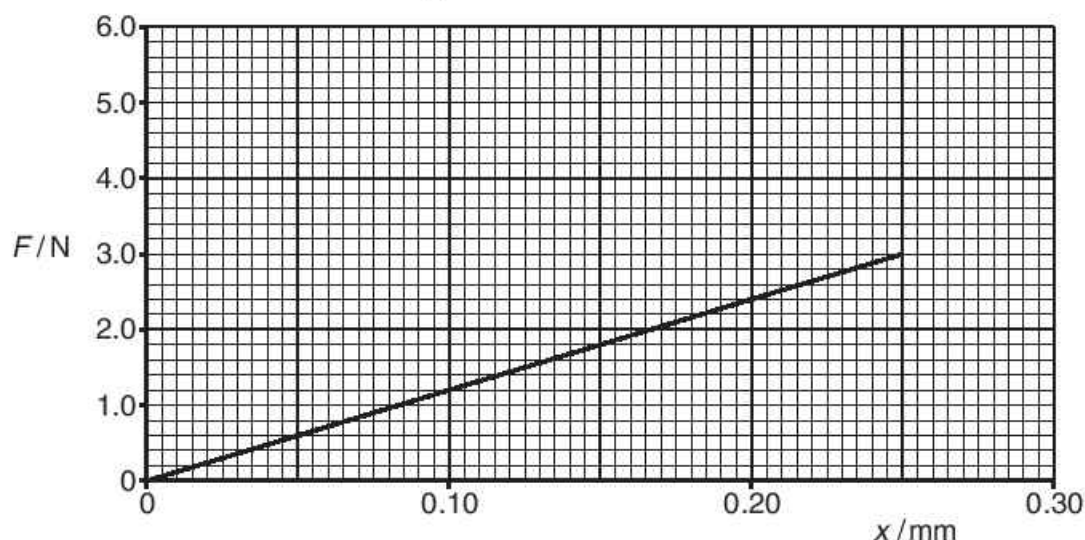


Fig. 5.1

- (i) State two quantities, other than the gradient of the graph in Fig. 5.1, that are required in order to determine the Young modulus of steel.

1.
 2. [1]

- (ii) Describe how the quantities you listed in (i) may be measured.

.....
 [2]

- (iii) A load of 3.0 N is applied to the wire. Use Fig. 5.1 to calculate the energy stored in the wire.

energy = J [2]

- (c) A copper wire has the same original dimensions as the steel wire. The Young modulus for steel is $2.2 \times 10^{11} \text{ N m}^{-2}$ and for copper is $1.1 \times 10^{11} \text{ N m}^{-2}$.

On Fig. 5.1, sketch the variation with x of F for the copper wire for extensions up to 0.25 mm. The copper wire is not extended beyond its limit of proportionality.

[2]

13 (a) Explain what is meant by *plastic deformation*.

9702/21/O/N/12/Q5

.....
..... [1]

(b) A copper wire of uniform cross-sectional area $1.54 \times 10^{-6} \text{ m}^2$ and length 1.75 m has a breaking stress of $2.20 \times 10^8 \text{ Pa}$. The Young modulus of copper is $1.20 \times 10^{11} \text{ Pa}$.

(i) Calculate the breaking force of the wire.

breaking force = N [2]

(ii) A stress of $9.0 \times 10^7 \text{ Pa}$ is applied to the wire. Calculate the extension.

extension = m [2]

(c) Explain why it is not appropriate to use the Young modulus to determine the extension when the breaking force is applied.

.....
..... [1]

.....
 [1]

- (b) A spring is attached to a support and hangs vertically, as shown in Fig. 6.1. An object M of mass 0.41 kg is attached to the lower end of the spring. The spring extends until M is at rest at R.

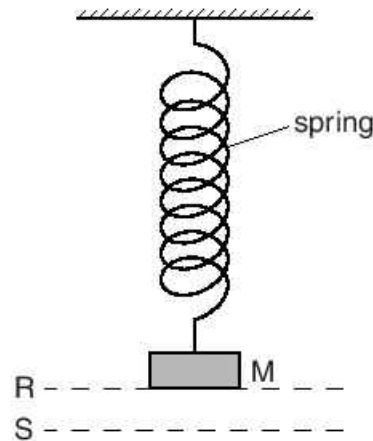


Fig. 6.1

The spring constant of the spring is 25 N m^{-1} . Show that the extension of the spring is about 0.16 m.

[2]

- (c) The object M in Fig. 6.1 is pulled down a further 0.060 m to S and is then released.

For M, just as it is released,

- (i) state the forces acting on M,

..... [1]

- (ii) calculate the acceleration of M.

acceleration = ms^{-2} [3]

- (d) Describe and explain the energy changes from the time the object M in Fig. 6.1 is released to the time it first returns to R.

.....

 [2]

15 Energy is stored in a metal wire that is extended elastically.

9702/21/M/J/13/Q1

(a) Explain what is meant by *extended elastically*.

.....
..... [2]

(b) Show that the SI units of energy per unit volume are $\text{kg m}^{-1} \text{s}^{-2}$.

[2]

(c) For a wire extended elastically, the elastic energy per unit volume X is given by

$$X = C\varepsilon^2 E$$

where C is a constant,

ε is the strain of the wire,

and E is the Young modulus of the wire.

Show that C has no units.

[3]

16 (a) Define

9702/23/M/J/13/Q4

(i) stress,

..... [1]

(ii) strain.

..... [1]

(b) The Young modulus of the metal of a wire is 0.17 TPa. The cross-sectional area of the wire is 0.18 mm^2 .

The wire is extended by a force F . This causes the length of the wire to be increased by 0.095%.

Calculate

(i) the stress,

stress = Pa [4]

(ii) the force F .

$F =$ N [2]

17 (a) Define the *Young modulus*.

9702/22/M/J/14/Q5

.....
..... [1]

- (b) Two wires P and Q of the same material and same original length l_0 are fixed so that they hang vertically, as shown in Fig. 5.1.

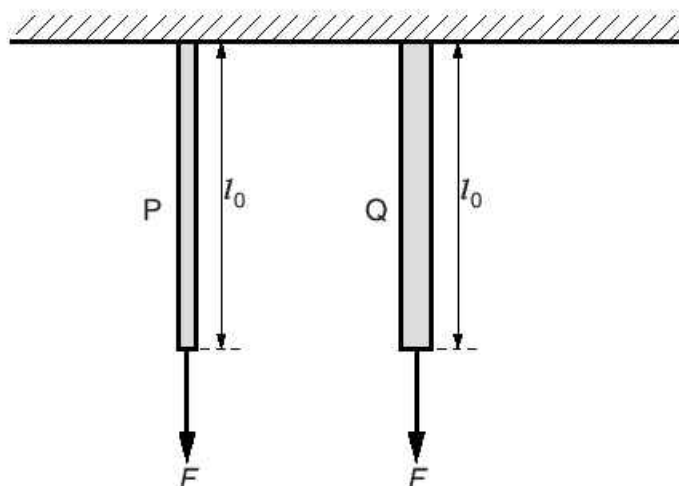


Fig. 5.1 (not to scale)

The diameter of P is d and the diameter of Q is $2d$. The same force F is applied to the lower end of each wire.

Show your working and determine the ratio

(i) $\frac{\text{stress in P}}{\text{stress in Q}},$

ratio = [2]

(ii) $\frac{\text{strain in P}}{\text{strain in Q}}.$

ratio = [2]

18 A spring hangs vertically from a point P, as shown in Fig. 4.1.

9702/23/M/J/14/Q4

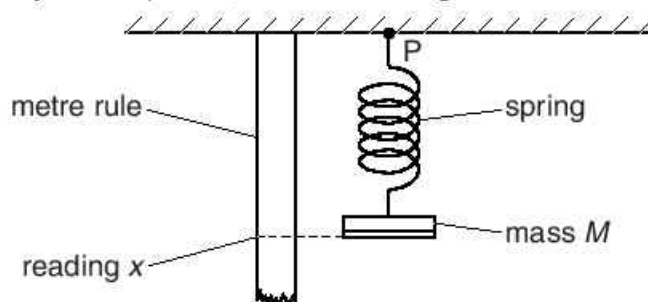


Fig. 4.1

A mass M is attached to the lower end of the spring. The reading x from the metre rule is taken, as shown in Fig. 4.1. Fig. 4.2 shows the relationship between x and M .

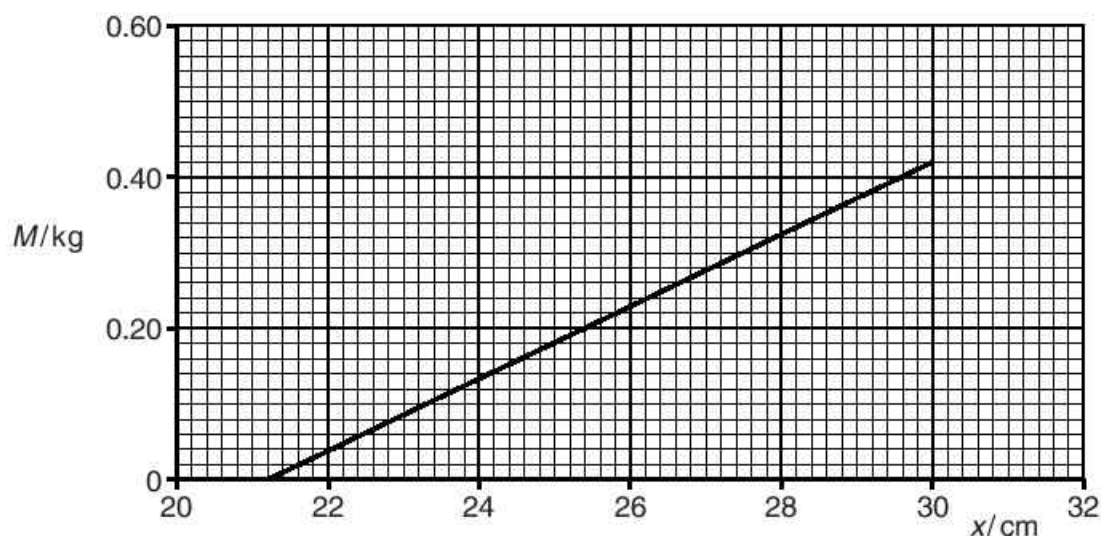


Fig. 4.2

(a) Explain how the apparatus in Fig. 4.1 may be used to determine the load on the spring at the elastic limit.

.....

 [2]

(b) State and explain whether Fig. 4.2 suggests that the spring obeys Hooke's law.

.....

 [2]

(c) Use Fig. 4.2 to determine the spring constant, in Nm^{-1} , of the spring.

spring constant = Nm^{-1} [3]

19 (a) Compare the molecular motion of a liquid with

9702/22/O/N/14/Q4

(i) a solid,

.....
.....[2]

(ii) a gas.

.....
.....[1]

(b) (i) A ductile material in the form of a wire is stretched up to its breaking point. On Fig. 4.1, sketch the variation with extension x of the stretching force F .

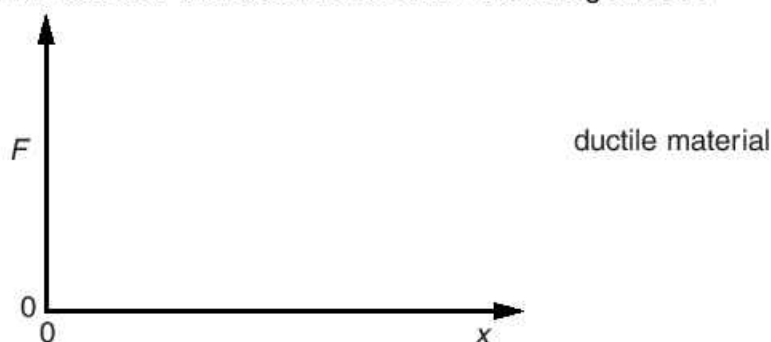


Fig. 4.1

[1]

(ii) On Fig. 4.2, sketch the variation with extension x of the stretching force F for a brittle material up to its breaking point.

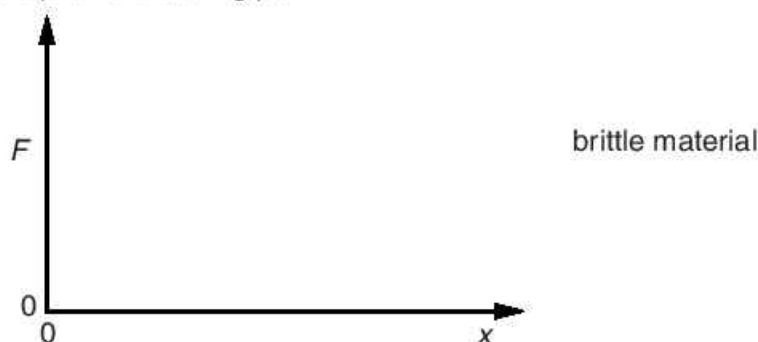


Fig. 4.2

[1]

(c) Describe a similarity and a difference between ductile and brittle materials.

similarity:
.....
difference:
.....

[2]

- 20 A spring is kept horizontal by attaching it to points A and B, as shown in Fig. 4.1.

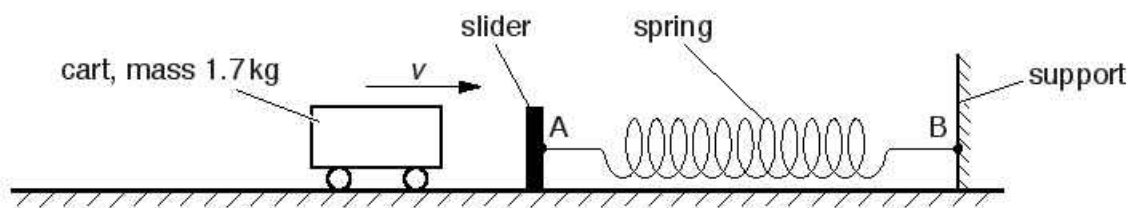


Fig. 4.1

Point A is on a movable slider and point B is on a fixed support. A cart of mass 1.7 kg has horizontal velocity v towards the slider. The cart collides with the slider. The spring is compressed as the cart comes to rest. The variation of compression x of the spring with force F exerted on the spring is shown in Fig. 4.2.

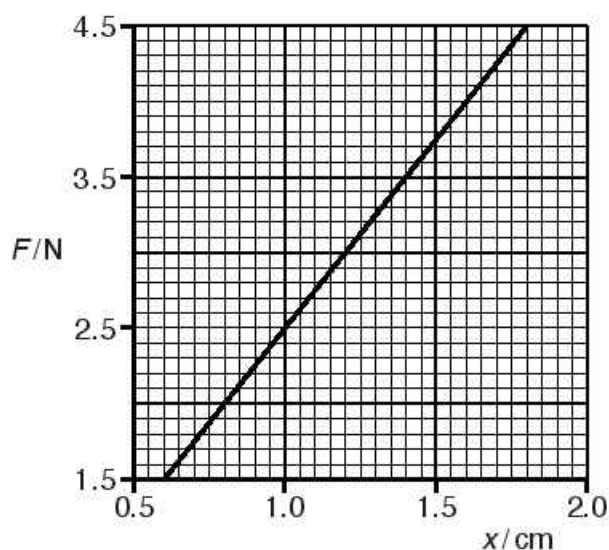


Fig. 4.2

Fig. 4.2 shows the compression of the spring for $F = 1.5\text{ N}$ to $F = 4.5\text{ N}$. The cart comes to rest when F is 4.5 N.

- (a) Use Fig. 4.2 to

- (i) show that the compression of the spring obeys Hooke's law,

.....
 [2]

- (ii) determine the spring constant of the spring,

spring constant = Nm^{-1} [2]

- (iii) determine the elastic potential energy E_p stored in the spring due to the cart being brought to rest.

$$E_p = \dots\dots\dots \text{ J [3]}$$

- (b) Calculate the speed v of the cart as it makes contact with the slider. Assume that all the kinetic energy of the cart is converted to the elastic potential energy of the spring.

$$\text{speed} = \dots\dots\dots \text{ m s}^{-1} \text{ [2]}$$

21 (a) Define the *Young modulus*.

.....
[1]

(b) The Young modulus of steel is 1.9×10^{11} Pa. The Young modulus of copper is 1.2×10^{11} Pa.

A steel wire and a copper wire each have the same cross-sectional area and length. The two wires are each extended by equal forces.

(i) Use the definition of the Young modulus to determine the ratio

$$\frac{\text{extension of the copper wire}}{\text{extension of the steel wire}}.$$

ratio =[3]

(ii) The two wires are each extended by a force. Both wires obey Hooke's law.

On Fig. 3.1, sketch a graph for each wire to show the variation with extension of the force.

Label the line for steel with the letter **S** and the line for copper with the letter **C**.



Fig. 3.1

[1]

22 (a) State Hooke's law.

.....
 [1]

(b) The variation with compression x of the force F acting on a spring is shown in Fig. 3.1.

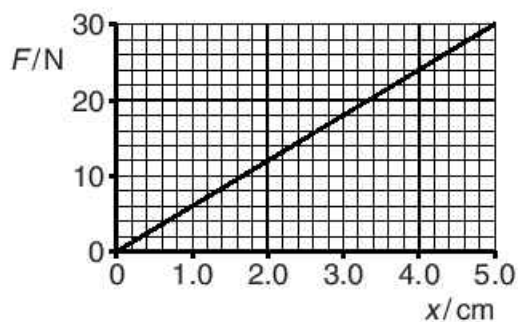


Fig. 3.1

The spring is fixed to the closed end of a horizontal tube. A block is pushed into the tube so that the spring is compressed, as shown in Fig. 3.2.

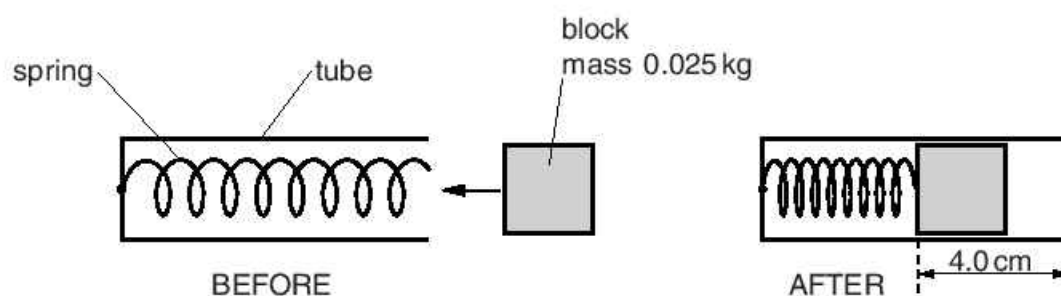


Fig. 3.2 (not to scale)

The compression of the spring is 4.0 cm. The mass of the block is 0.025 kg.

(i) Calculate the spring constant of the spring.

spring constant = Nm^{-1} [2]

(ii) Show that the work done to compress the spring by 4.0 cm is 0.48 J.

(iii) The block is now released and accelerates along the tube as the spring returns to its original length. The block leaves the end of the tube with a speed of 6.0 m s^{-1} . [2]

1. Calculate the kinetic energy of the block as it leaves the end of the tube.

kinetic energy = J [2]

2. Assume that the spring has negligible kinetic energy as the block leaves the tube. Determine the average resistive force acting against the block as it moves along the tube.

resistive force = N [3]

(iv) Determine the efficiency of the transfer of elastic potential energy from the spring to the kinetic energy of the block.

efficiency = [2]

- 23 The Young modulus of the material of a wire can be determined using the apparatus shown in Fig. 3.1.

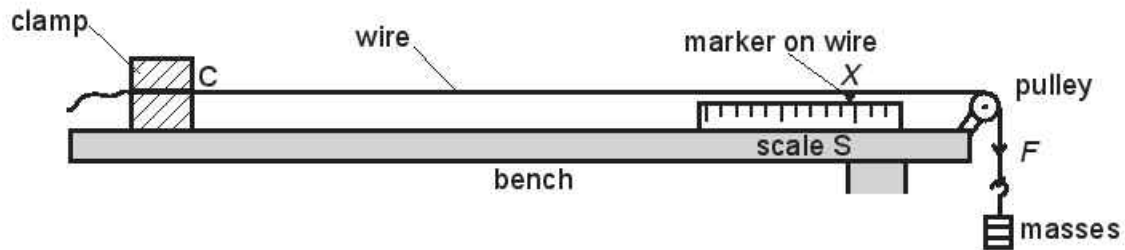


Fig. 3.1

One end of the wire is clamped at C and a marker is attached to the wire above a scale S. A force to extend the wire is applied by attaching masses to the other end of the wire.

The reading X of the marker on the scale S is determined for different forces F applied to the end of the wire. The variation with X of F is shown in Fig. 3.2.

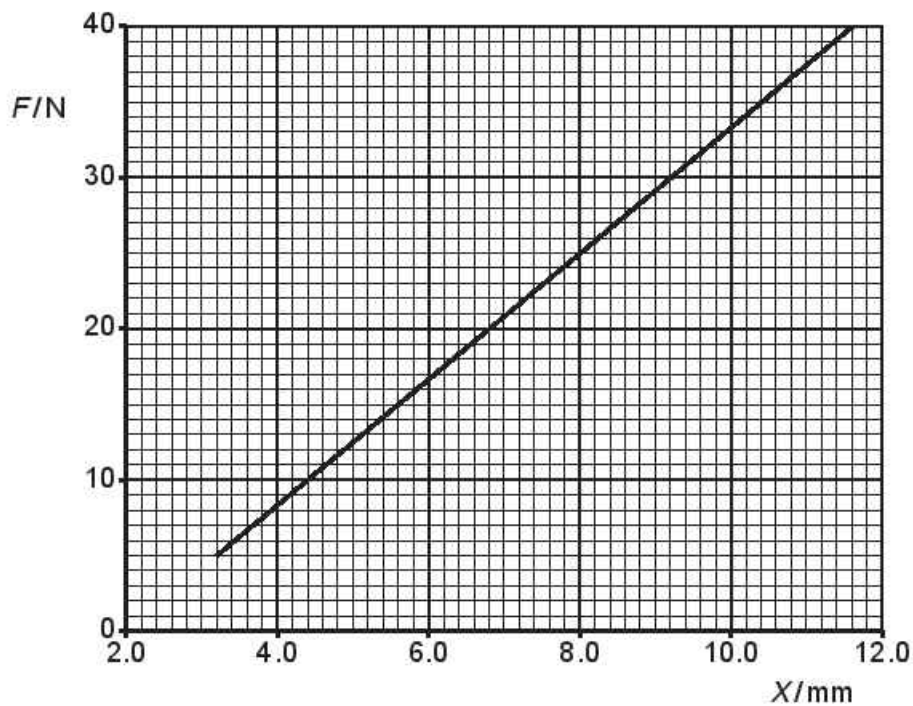


Fig. 3.2

- (a) The length of the wire from C to the marker for $F = 0$ is 3.50m. The diameter of the wire is 0.38mm.

Use the gradient of the line in Fig. 3.2 to determine the Young modulus E of the material of the wire in TPa.

$E = \dots\dots\dots$ TPa [3]

- (b) The experiment is repeated with a thicker wire of the same material and length.

State how the range of the force F must be changed to obtain the same range of scale readings as in Fig. 3.2.

.....
.....
..... [1]

24 (a) Define *strain*.

.....
 [1]

- (b) A wire is designed to ensure that its strain does not exceed 4.0×10^{-4} when a force of 8.0 kN is applied. The Young modulus of the metal of the wire is 2.1×10^{11} Pa. It may be assumed that the wire obeys Hooke's law.

For a force of 8.0 kN, calculate, for the wire,

- (i) the maximum stress,

maximum stress = Pa [2]

- (ii) the minimum cross-sectional area.

minimum cross-sectional area = m^2 [2]

- 25 A spring is attached at one end to a fixed point and hangs vertically with a cube attached to the other end. The cube is initially held so that the spring has zero extension, as shown in Fig. 3.1.

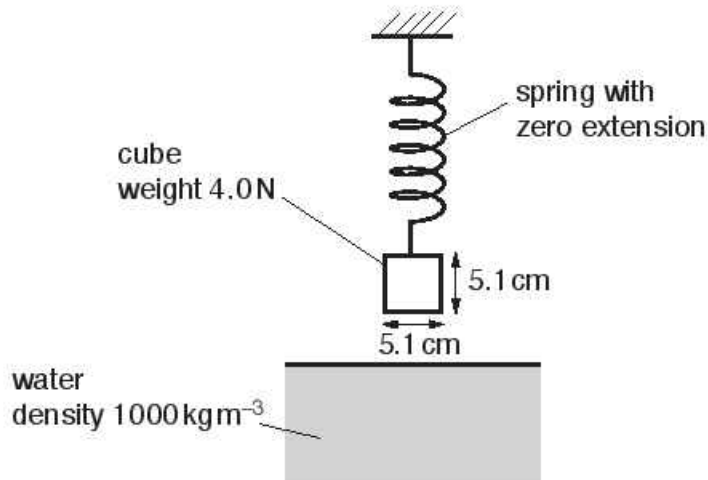


Fig. 3.1

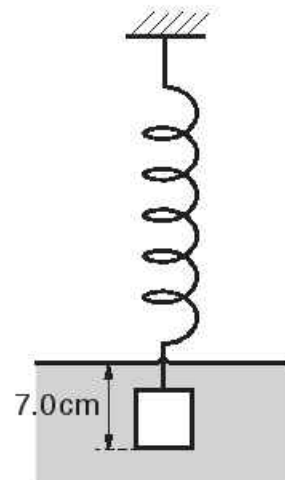


Fig. 3.2

The cube has weight 4.0 N and sides of length 5.1 cm. The cube is released and sinks into water as the spring extends. The cube reaches equilibrium with its base at a depth of 7.0 cm below the water surface, as shown in Fig. 3.2.

The density of the water is 1000 kg m⁻³.

- (a) Calculate the difference in the pressure exerted by the water on the bottom face and on the top face of the cube.

difference in pressure = Pa [2]

- (b) Use your answer in (a) to show that the upthrust on the cube is 1.3 N.

- (c) Calculate the force exerted on the spring by the cube when it is in equilibrium in the water.

force = N [1]

- (d) The spring obeys Hooke's law and has a spring constant of 30 N m^{-1} .

Determine the initial height above the water surface of the base of the cube before it was released.

height above surface = cm [3]

- (e) The cube in the water is released from the spring.

- (i) Determine the initial acceleration of the cube.

acceleration = ms^{-2} [2]

- (ii) Describe and explain the variation, if any, of the acceleration of the cube as it sinks in the water.

.....
.....
..... [2]

- 26** A solid cylinder is lifted out of oil by a wire attached to a motor. Fig. 5.1 shows two different positions X and Y of the cylinder during the lifting process.

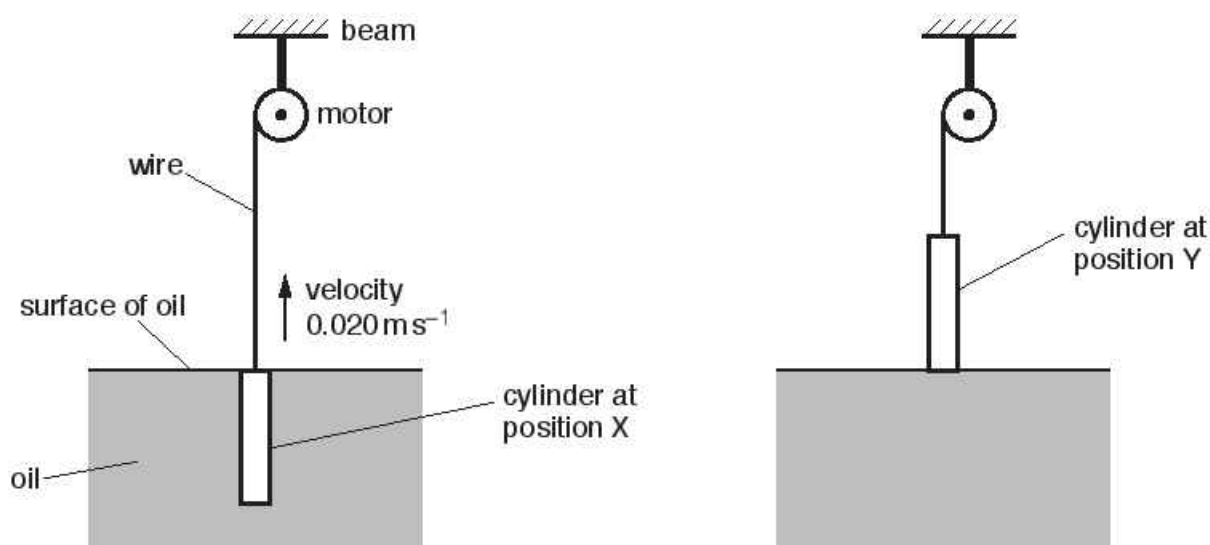


Fig. 5.1

The motor is fixed to an overhead beam.

The cylinder has cross-sectional area 0.018 m^2 , length 1.2 m and weight 560 N .

The density of the oil is 940 kg m^{-3} .

Throughout the lifting process, the cylinder moves vertically upwards with a constant velocity of 0.020 m s^{-1} . The viscous force of the oil acting on the cylinder is negligible.

- (a)** Calculate the density of the cylinder.

density = kg m^{-3} [2]

- (b)** For the cylinder at position X, show that the upthrust due to the oil is 200 N .

(c) Calculate, for the moving cylinder at position X,

(i) the tension in the wire,

tension = N [1]

(ii) the power output of the motor.

power = W [2]

(d) The cylinder is raised with constant velocity from position X to position Y.

(i) State and explain the variation, if any, of the power output of the motor as the cylinder is raised. Numerical values are not required.

.....
.....
.....
.....
.....[3]

(ii) The rate of energy output of the motor is less than the rate of increase of gravitational potential energy of the cylinder. Without calculation, explain this difference.

.....
.....[1]

27 (a) Define the *Young modulus* of a material.

.....
 [1]

(b) A metal rod is compressed, as shown in Fig. 4.1.

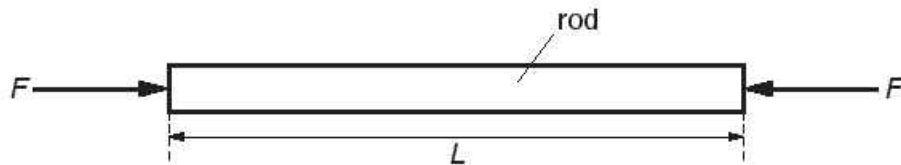


Fig. 4.1

The variation with compressive force F of the length L of the rod is shown in Fig. 4.2.

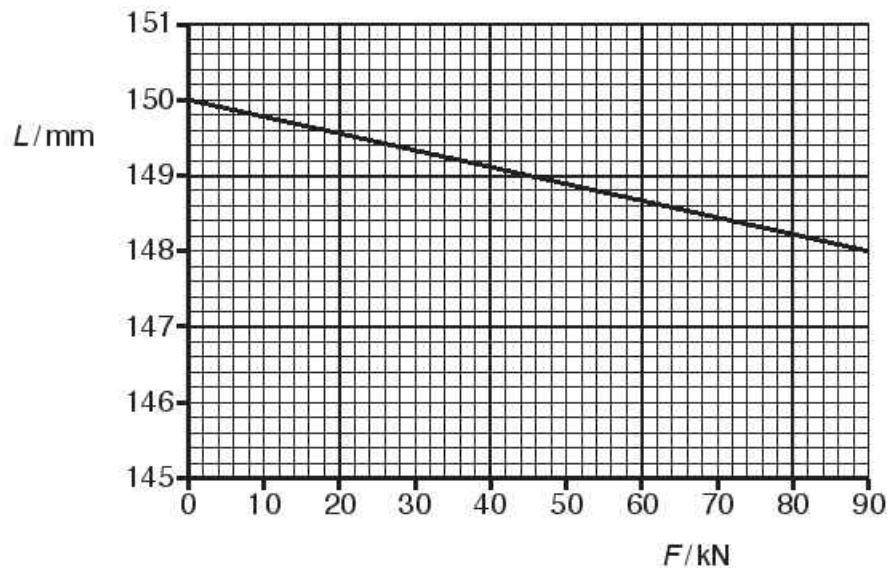


Fig. 4.2

Use Fig. 4.2 to

(i) determine the spring constant k of the rod,

$k = \dots\dots\dots \text{N m}^{-1}$ [2]

- (ii) determine the strain energy stored in the rod for $F = 90\text{ kN}$.

strain energy = J [3]

- (c) The rod in (b) has cross-sectional area A and is made of metal of Young modulus E . It is now replaced by a new rod of the same original length. The new rod has cross-sectional area $A/3$ and is made of metal of Young modulus $2E$. The compression of the new rod obeys Hooke's law.

On Fig. 4.2, sketch the variation with F of the length L for the new rod from $F = 0$ to $F = 90\text{ kN}$. [2]

9702/23/O/N/18/Q1

- 28 (a) Mass, length and time are all SI base quantities.

State two other SI base quantities.

1.

2.

[2]

- (b) A wire hangs between two fixed points, as shown in Fig. 1.1.

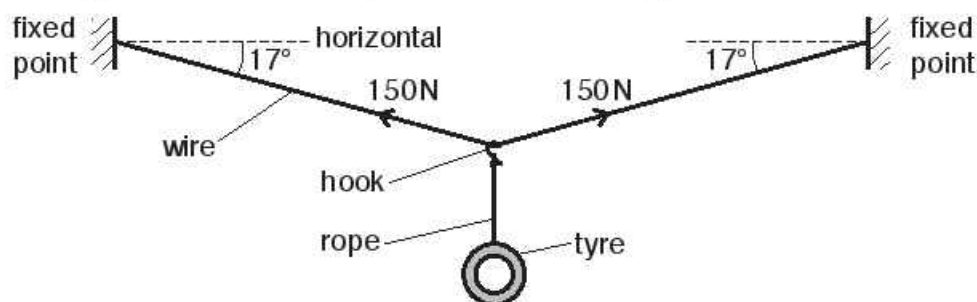


Fig. 1.1 (not to scale)

A child's swing is made by connecting a car tyre to the wire using a rope and a hook. The system is in equilibrium with the wire hanging at an angle of 17° to the horizontal. The tension in the wire is 150 N . Assume that the rope and hook have negligible weight.

- (i) Determine the weight of the tyre.

weight = N [2]

- (ii) The wire has a cross-sectional area of 7.5 mm^2 and is made of metal of Young modulus $2.1 \times 10^{11} \text{ Pa}$. The wire obeys Hooke's law.

Calculate, for the wire,

1. the stress,

stress = Pa [2]

2. the strain.

strain = [2]

9702/23/O/N/19/Q4

- 29 A ball X moves along a horizontal frictionless surface and collides with another ball Y, as illustrated in Fig. 4.1.

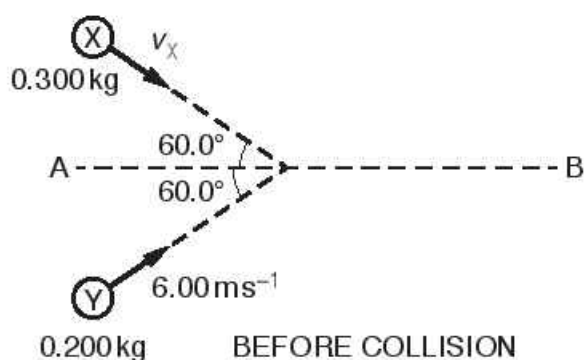


Fig. 4.1 (not to scale)

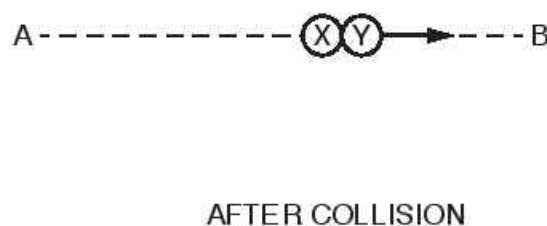


Fig. 4.2 (not to scale)

Ball X has mass 0.300 kg and initial velocity v_X at an angle of 60.0° to line AB.

Ball Y has mass 0.200 kg and initial velocity 6.00 ms^{-1} at an angle of 60.0° to line AB.

The balls stick together during the collision and then travel along line AB, as illustrated in Fig. 4.2.

- (a) (i) Calculate, to three significant figures, the component of the initial momentum of ball Y that is perpendicular to line AB.

component of momentum = kgms^{-1} [2]

- (ii) By considering the component of the initial momentum of each ball perpendicular to line AB, calculate, to three significant figures, v_X .

$v_X = \dots\dots\dots \text{ms}^{-1}$ [1]

- (iii) Show that the speed of the two balls after the collision is 2.4 ms^{-1} .

- (b) The two balls continue moving together along the horizontal frictionless surface towards a spring, as illustrated in Fig. 4.3.

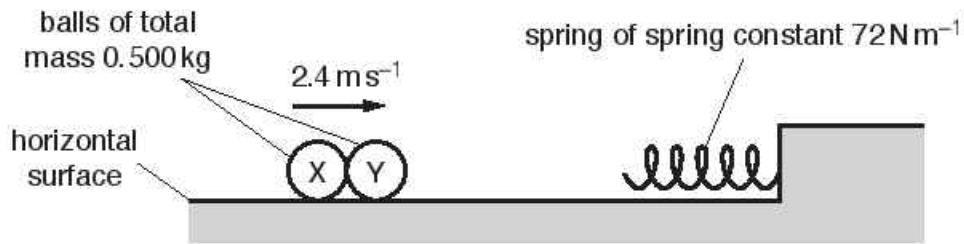


Fig. 4.3

The balls hit the spring and remain stuck together as they decelerate to rest. All the kinetic energy of the balls is converted into elastic potential energy of the spring. The energy E stored in the spring is given by

$$E = \frac{1}{2}kx^2$$

where k is the spring constant of the spring and x is its compression.
The spring obeys Hooke's law and has a spring constant of 72 N m^{-1} .

- (i) Determine the maximum compression of the spring caused by the two balls.

maximum compression = m [3]

(ii) On Fig. 4.4, sketch graphs to show the variation with compression x of the spring, from zero to maximum compression, of:

1. the magnitude of the deceleration a of the balls
2. the kinetic energy E_k of the balls.

Numerical values are not required.

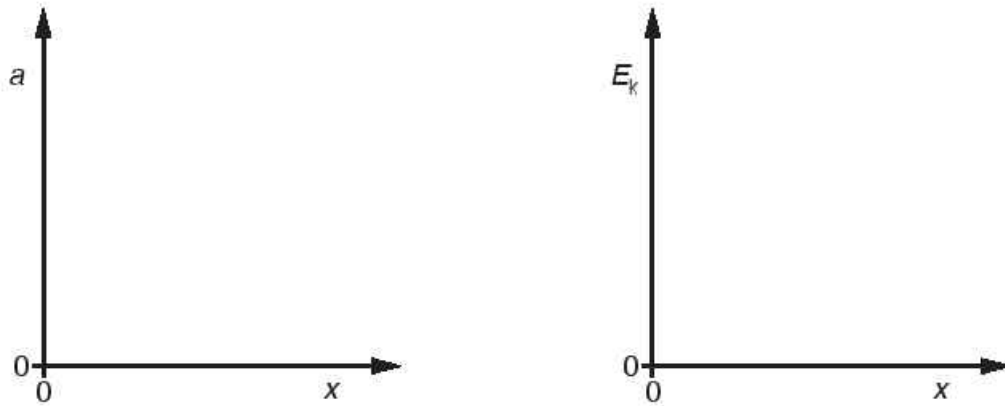


Fig. 4.4

[3]

- 30 One end of a wire is attached to a fixed point. A force F is applied to the wire to cause extension x . The variation with F of x is shown in Fig. 5.1.

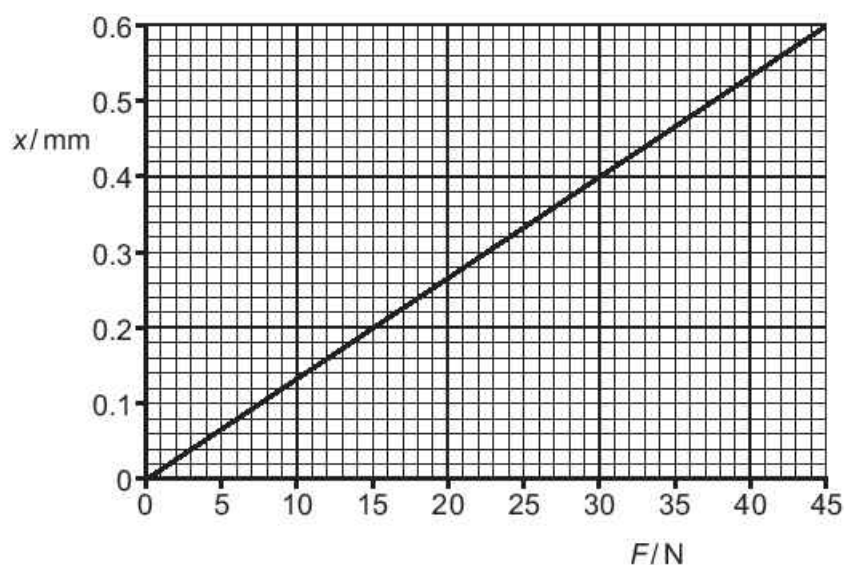


Fig. 5.1

The wire has a cross-sectional area of $4.1 \times 10^{-7} \text{ m}^2$ and is made of metal of Young modulus $1.7 \times 10^{11} \text{ Pa}$. Assume that the cross-sectional area of the wire remains constant as the wire extends.

- (a) State the name of the law that describes the relationship between F and x shown in Fig. 5.1.

..... [1]

- (b) The wire has an extension of 0.48 mm.

Determine:

- (i) the stress

stress = Pa [2]

- (ii) the strain.

strain = [2]

- (c) The resistivity of the metal of the wire is $3.7 \times 10^{-7} \Omega \text{ m}$.

Determine the change in resistance of the wire when the extension x of the wire changes from $x = 0.48 \text{ mm}$ to $x = 0.60 \text{ mm}$.

change in resistance = Ω [3]

- (d) A force of greater than 45 N is now applied to the wire.

Describe how it may be checked that the elastic limit of the wire has not been exceeded.

.....
..... [1]

- 31 (a) State the principle of moments.

.....

.....

..... [2]

- (b) In a bicycle shop, two wheels hang from a horizontal uniform rod AC, as shown in Fig. 3.1.

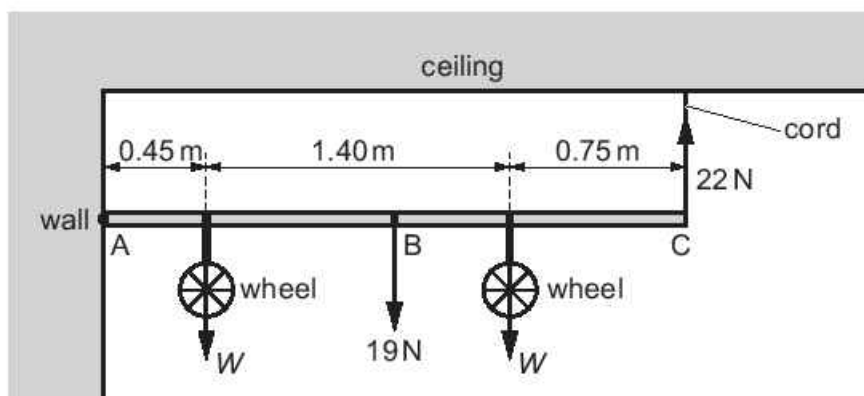


Fig. 3.1 (not to scale)

The rod has weight 19 N and is freely hinged to a wall at end A. The other end C of the rod is attached by a vertical elastic cord to the ceiling. The centre of gravity of the rod is at point B. The weight of each wheel is W and the tension in the cord is 22 N.

- (i) By taking moments about end A, show that the weight W of each wheel is 14 N.

[2]

- (ii) Determine the magnitude and the direction of the force acting on the rod at end A.

magnitude = N

direction

[2]

- (c) The unstretched length of the cord in (b) is 0.25 m. The variation with length L of the tension F in the cord is shown in Fig. 3.2.

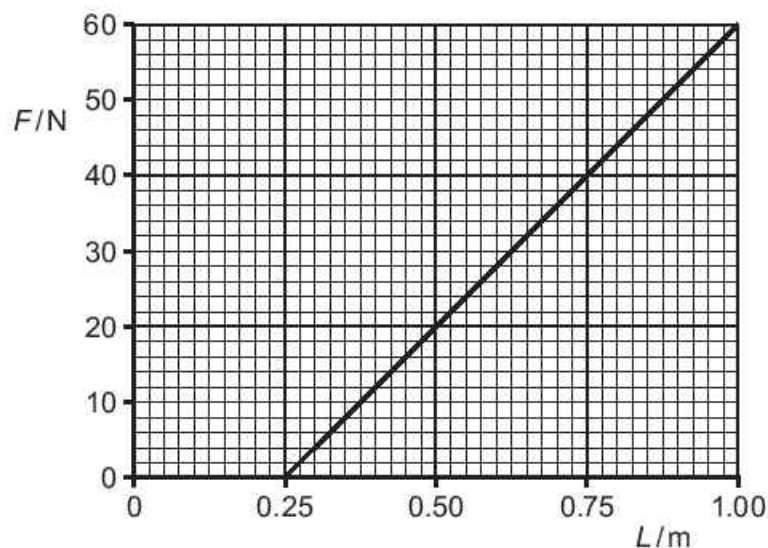


Fig. 3.2

- (i) State and explain whether Fig. 3.2 suggests that the cord obeys Hooke's law.

.....

 [2]

- (ii) Calculate the spring constant k of the cord.

$k = \dots\dots\dots \text{Nm}^{-1}$ [2]

- (iii) On Fig. 3.2, shade the area that represents the work done to extend the cord when the tension is increased from $F = 0$ to $F = 40$ N. [1]

32. 9702/21/O/N/20/Q4

(a) Define, for a wire:

(i) *stress*

.....
 [1]

(ii) *strain*.

.....
 [1]

(b) (i) A school experiment is performed on a metal wire to determine the Young modulus of the metal. A force is applied to one end of the wire which is fixed at the other end. The variation of the force F with extension x of the wire is shown in Fig. 4.1.

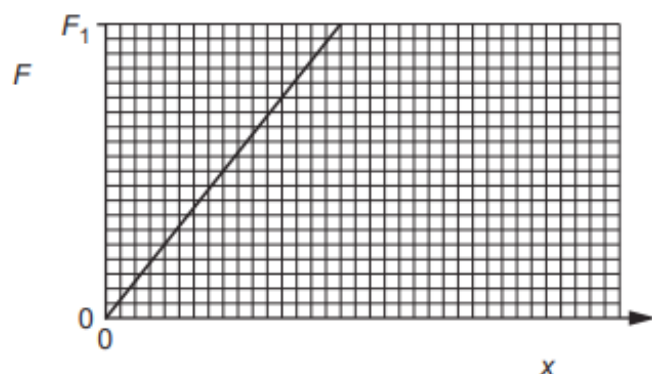


Fig. 4.1

The maximum force applied to the wire is F_1 .

The gradient of the graph line in Fig. 4.1 is G . The wire has initial length L and cross-sectional area A .

Determine an expression, in terms of A , G and L , for the Young modulus E of the metal.

$E =$ [2]

- (ii) A student repeats the experiment in **(b)(i)** using a new wire that has twice the diameter of the first wire. The initial length of the wire and the metal of the wire are unchanged.

On Fig. 4.1, draw the graph line representing the new wire for the force increasing from $F = 0$ to $F = F_1$. [2]

- (iii) Another student repeats the original experiment in **(b)(i)**, increasing the force beyond F_1 to a new maximum force F_2 . The new graph obtained is shown in Fig. 4.2.

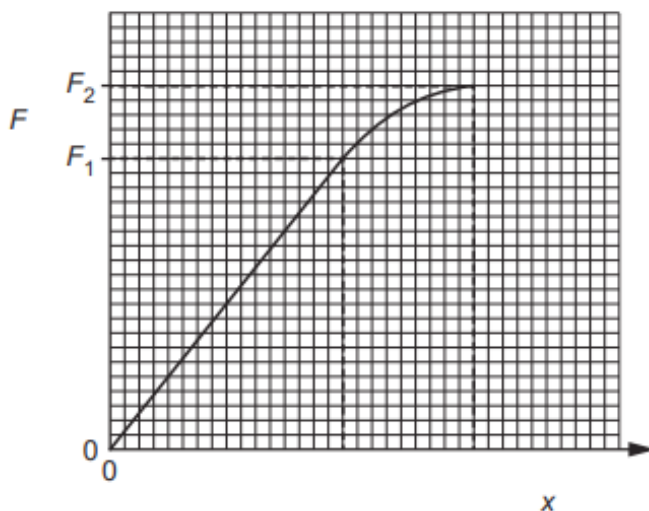


Fig. 4.2

- On Fig. 4.2, shade an area that represents the work done to extend the wire when the force is increased from F_1 to F_2 . [1]
- Explain how the student can check that the elastic limit of the wire was not exceeded when force F_2 was applied.

.....

 [1]

- (iv) Each student in the class performs the experiment in **(b)(i)**. The teacher describes the values of the Young modulus calculated by the students as having high accuracy and low precision.

Explain what is meant by *low precision*.

.....
 [1]

[Total: 9]

33. 9702/22/O/N/20/Q3

- (a) A spring is fixed at one end and is compressed by applying a force to the other end. The variation of the force F acting on the spring with its compression x is shown in Fig. 3.1.

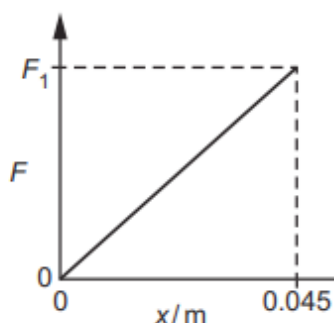


Fig. 3.1

A compression of 0.045 m is produced when a force F_1 acts on the spring. The spring has a spring constant of 800 N m^{-1} .

- (i) Determine F_1 .

$$F_1 = \dots\dots\dots \text{ N [2]}$$

- (ii) Use Fig. 3.1 to show that, for a compression of 0.045 m, the elastic potential energy of the spring is 0.81 J.

[2]

- (b) A child's toy uses the spring in (a) to launch a ball of mass 0.020 kg vertically into the air. The ball is initially held against one end of the spring which has a compression of 0.045 m. The spring is then released to launch the ball. The kinetic energy of the ball as it leaves the toy is 0.72 J.

- (i) The toy converts the elastic potential energy of the spring into the kinetic energy of the ball. Use the information in (a)(ii) to calculate the percentage efficiency of this conversion.

$$\text{efficiency} = \dots\dots\dots \% [1]$$

- (ii) Determine the initial momentum of the ball as it leaves the toy.

momentum = N s [3]

- (c) The ball in (b) leaves the toy at point A and moves vertically upwards through the air. Point B is the position of the ball when it is at maximum height h above point A, as illustrated in Fig. 3.2.

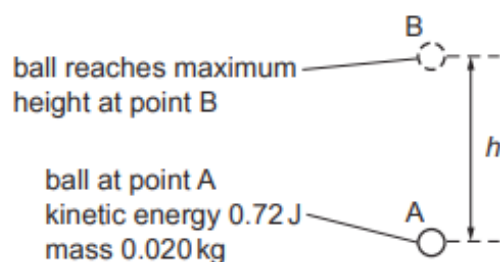


Fig. 3.2 (not to scale)

The gravitational potential energy of the ball increases by 0.60 J as it moves from A to B.

- (i) Calculate h .

h = m [2]

- (ii) Determine the average force due to air resistance acting on the ball for its movement from A to B.

average force = N [2]

(iii) When there is air resistance, the ball takes time T to move from A to B.

State and explain whether the time taken for the ball to move from A to its maximum height will be more than, less than or equal to time T if there is **no** air resistance.

.....

..... [1]

[Total: 13]

34. 9702/23/O/N/20/Q4

(a) State Hooke's law.

[1]

(b) A spring is fixed at one end. A compressive force F is applied to the other end. The variation of the force F with the compression x of the spring is shown in Fig. 4.1.

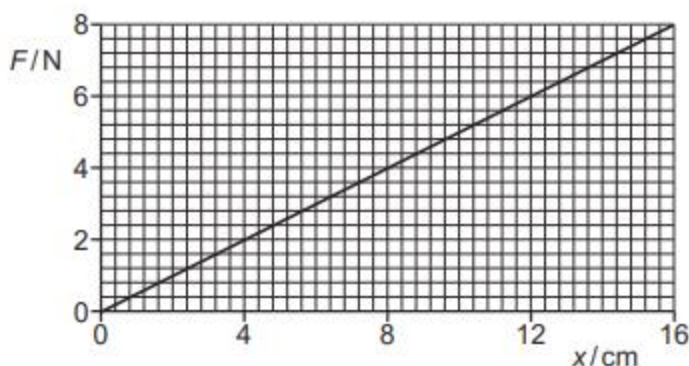


Fig. 4.1

Show that the elastic potential energy of the spring is 0.64 J when its compression is 16.0 cm.

[2]

(c) The spring in (b) is used to project a toy car along a track from point X to point Y, as illustrated in Fig. 4.2.

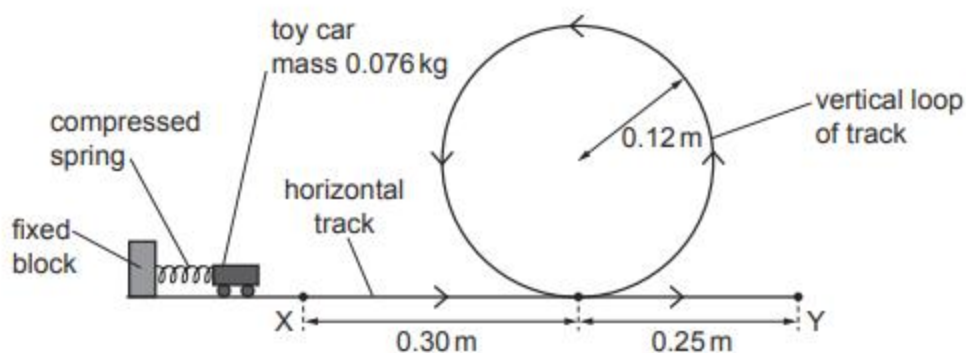


Fig. 4.2 (not to scale)

The spring is initially given a compression of 16.0 cm. The car of mass 0.076 kg is held against one end of the compressed spring. When the spring is released it projects the car forward. The car leaves the spring at point X with kinetic energy that is equal to the initial elastic potential energy of the compressed spring.

The car follows the track around a vertical loop of radius 0.12 m and then passes point Y. Assume that friction and air resistance are negligible.

Calculate:

- (i) the speed of the car at X

speed = ms^{-1} [2]

- (ii) the kinetic energy of the car when it is at the top of the loop

kinetic energy = J [3]

- (iii) the speed of the car at Y.

speed = ms^{-1} [1]

- (d) In practice, a resistive force due to friction and air resistance acts on the car so that its kinetic energy at Y is 0.23 J less than its kinetic energy at X.

Determine the average resistive force acting on the car for its movement from X to Y.

average resistive force = N [3]

[Total: 12]

35. 9702/22/F/M/21/Q3

A spring is extended by a force. The variation with extension x of the force F is shown in Fig. 3.1.

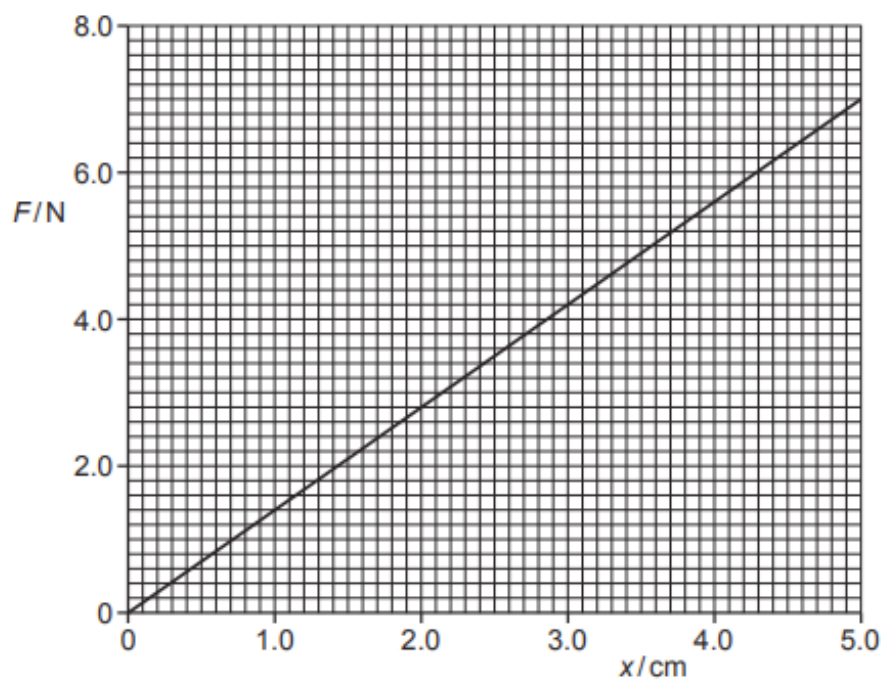


Fig. 3.1

(a) State the name of the law that relates the force and extension of the spring shown in Fig. 3.1.

..... [1]

(b) Determine:

(i) the spring constant, in Nm^{-1} , of the spring

spring constant = Nm^{-1} [2]

(ii) the strain energy (elastic potential energy) in the spring when the extension is 4.0 cm.

strain energy = J [2]

- (c) One end of the spring is attached to a fixed point. A cylinder that is submerged in a liquid is now suspended from the other end of the spring, as shown in Fig. 3.2.

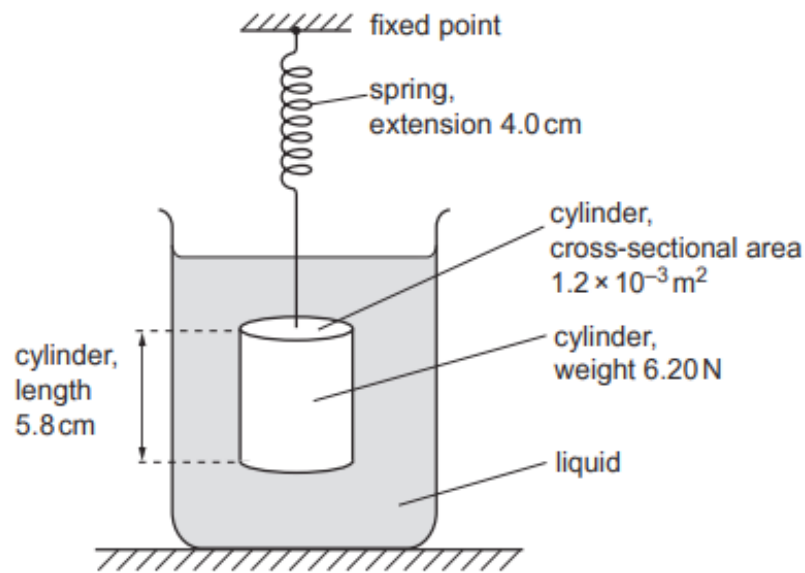


Fig. 3.2

The cylinder has length 5.8 cm, cross-sectional area $1.2 \times 10^{-3} \text{ m}^2$ and weight 6.20 N. The cylinder is in equilibrium when the extension of the spring is 4.0 cm.

- (i) Show that the upthrust acting on the cylinder is 0.60 N.

[1]

- (ii) Calculate the difference in pressure between the bottom face and the top face of the cylinder.

difference in pressure = Pa [2]

(iii) Calculate the density of the liquid.

density = kg m^{-3} [2]

(d) The liquid in (c) is replaced by another liquid of greater density.

State the effect, if any, of this change on:

(i) the upthrust acting on the cylinder

..... [1]

(ii) the extension of the spring.

..... [1]

[Total: 12]

36. 9702/23/M/J/21/Q3

(a) Define the *moment* of a force about a point.

.....

.....

..... [2]

(b) Fig. 3.1 shows a type of balance that is used for measuring mass.

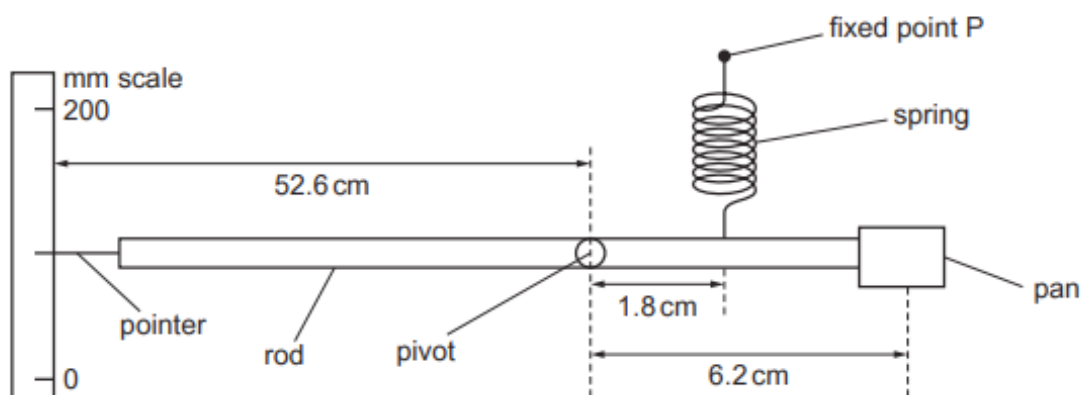


Fig. 3.1 (not to scale)

A rigid rod is pivoted about a point 6.2 cm from the centre of a pan which is attached to one end. The object being measured is placed on the centre of this pan.

A spring, attached to the rod 1.8 cm from the pivot, is attached at its other end to a fixed point P. The spring obeys Hooke's law over the full range of operation of the balance.

A pointer, on the other side of the pivot, is set against a millimetre scale which is a distance 52.6 cm from the pivot.

When the system is in equilibrium with no mass on the pan, the rod is horizontal and the pointer indicates a reading on the scale of 86 mm.

An object of mass 0.472 kg is now placed on the pan. As a result, the pointer moves to indicate a reading of 123 mm on the scale when the system is again in equilibrium.

(i) Show that the increase in the length of the spring is approximately 1.3 mm.

- (ii) Calculate the magnitude of the moment about the pivot of the weight of the object.

moment = N m [2]

- (iii) Use your answer in (b)(ii) to determine the increase in the tension in the spring due to the 0.472 kg mass.

increase in tension = N [2]

- (iv) Use the information in (b)(i) and your answer in (b)(iii) to determine the spring constant k of the spring. Give a unit with your answer.

k = unit [2]

[Total: 10]

37. 9702/21/O/N/21/Q3

- (a) A uniform metal bar, initially unstretched, has sides of length w , x and y , as shown in Fig. 3.1.

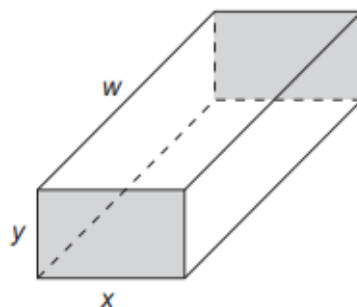


Fig. 3.1

The bar is now stretched by a tensile force F applied to the shaded ends. The changes in the lengths x and y are negligible. The bar now has sides of length x , y and z , as shown in Fig. 3.2.

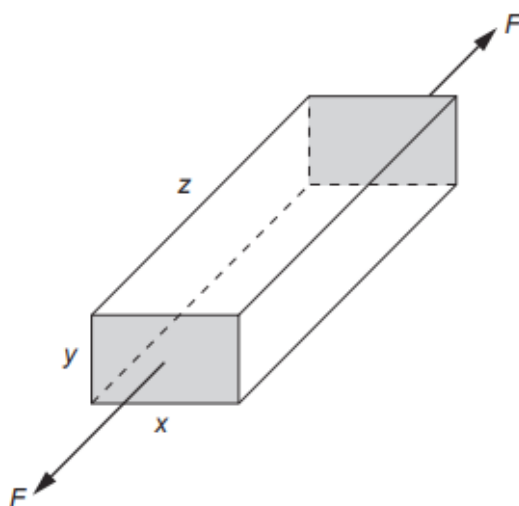


Fig. 3.2

Determine expressions, in terms of some or all of F , w , x , y and z , for:

- (i) the stress σ applied to the bar by the tensile force

$$\sigma = \dots\dots\dots [1]$$

- (ii) the strain ε in the bar due to the tensile force

$$\varepsilon = \dots\dots\dots [1]$$

- (iii) the Young modulus E of the metal from which the bar is made.

$$E = \dots\dots\dots [2]$$

- (b) A copper wire is stretched by a tensile force that gradually increases from 0 to 280 N. The variation with extension of the tensile force is shown in Fig. 3.3.

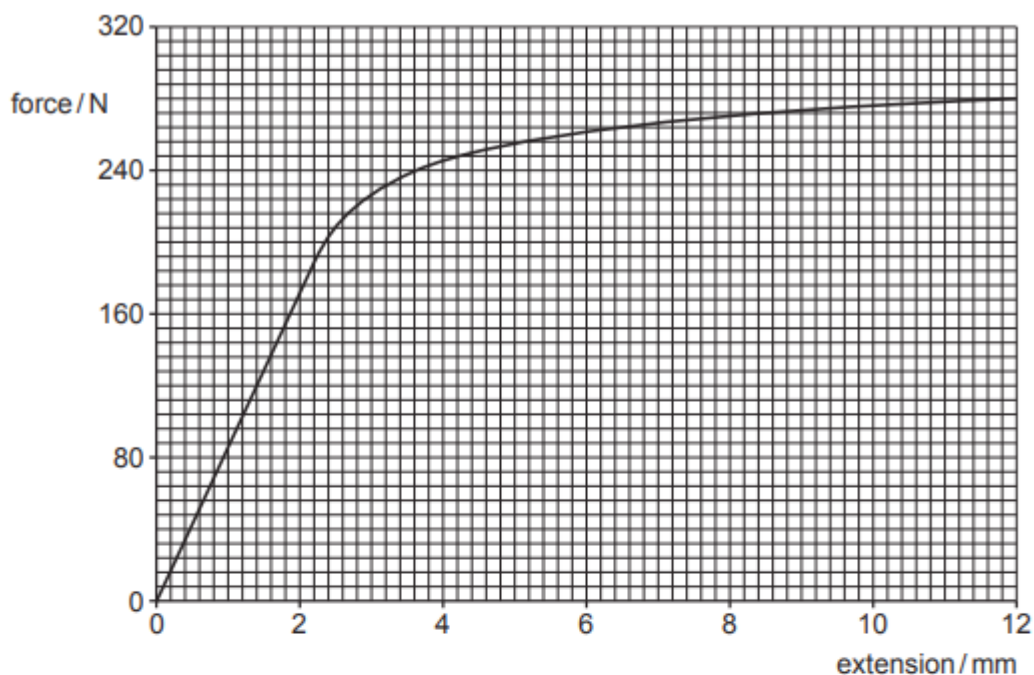


Fig. 3.3

- (i) State the maximum extension of the wire for which it obeys Hooke's law.

extension = mm [1]

- (ii) Use Fig. 3.3 to determine the strain energy in the wire when the tensile force is 120 N.

strain energy = J [3]

- (iii) Explain why the work done in stretching the wire to an extension of 12 mm is not equal to the energy recovered when the tensile force is removed.

.....

 [2]

[Total: 10]

38. 9702/21/M/J/22/Q4

A spring is suspended from a fixed point at one end and a vertical force is applied to the other end, as shown in Fig. 4.1.

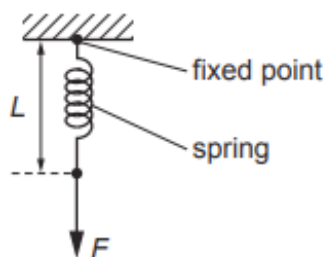


Fig. 4.1

The variation of the applied force F with the length L of the spring is shown in Fig. 4.2.

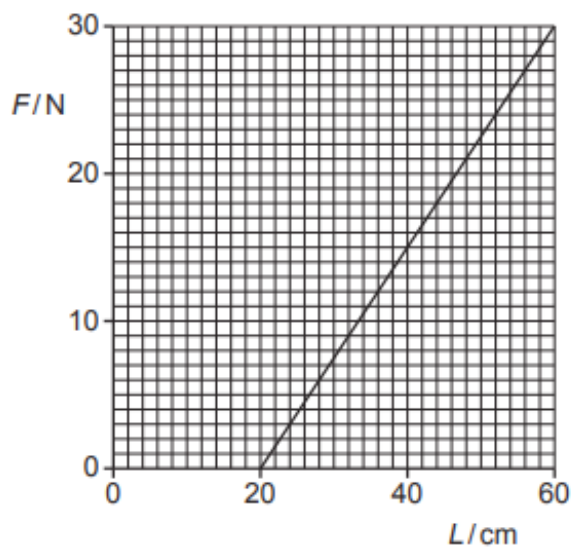


Fig. 4.2

(a) Determine the spring constant k of the spring.

$k = \dots\dots\dots \text{Nm}^{-1}$ [2]

(b) Determine the elastic potential energy in the spring when the applied force F is 15 N.

elastic potential energy = J [3]

[Total: 5]

39. 9702/21/O/N/22/Q2

A steel ball is projected horizontally from the top of a table, as shown in Fig. 2.1.

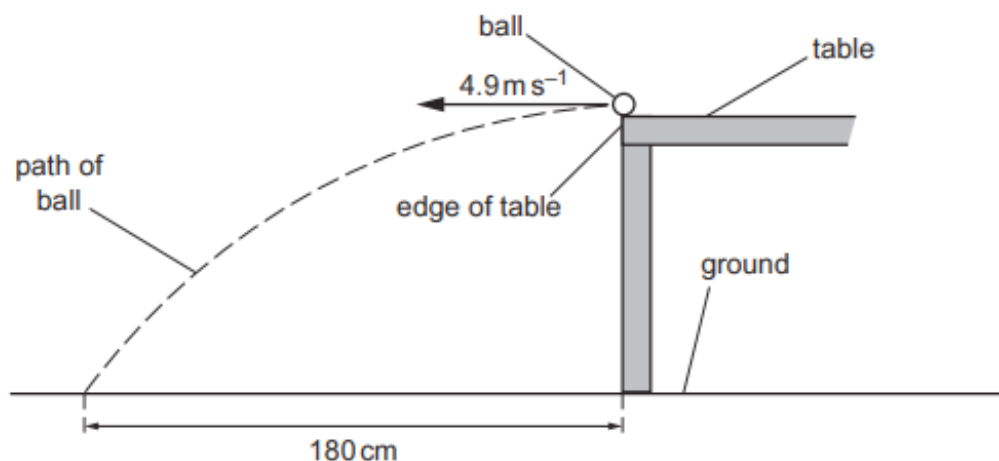


Fig. 2.1 (not to scale)

The ball is projected horizontally at a speed of 4.9 m s^{-1} . The ball lands on the ground a horizontal distance of 180 cm from the edge of the table.

Assume that air resistance is negligible.

- (a) (i)** Calculate the time taken for the ball to reach the ground.

time = s [1]

- (ii)** Calculate the vertical component of the velocity of the ball as it hits the ground.

velocity = m s^{-1} [2]

- (iii) Determine the magnitude and the angle to the horizontal of the velocity of the ball as it hits the ground.

magnitude of velocity = ms^{-1}

angle to the horizontal = $^{\circ}$
[3]

- (b) The ball is projected by means of a compressed spring which is attached to a fixed block as shown in Fig. 2.2.

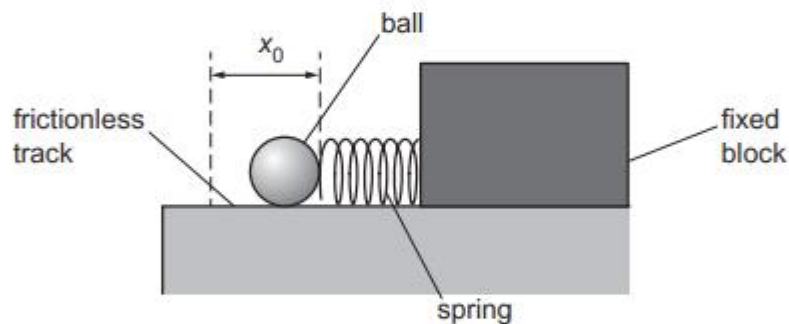


Fig. 2.2

The ball is placed on a frictionless track in front of the spring. The ball is then pulled back so that the spring has compression x_0 .

When the spring is released, the ball is projected horizontally as shown in Fig. 2.3.

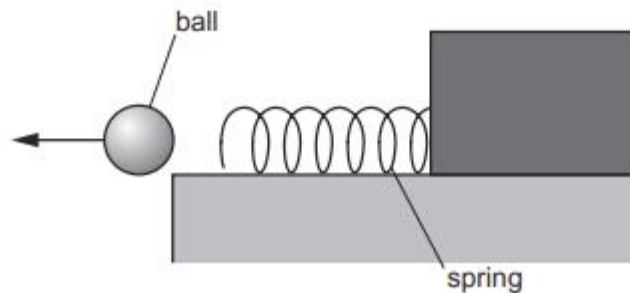


Fig. 2.3

The variation with compression x of the applied force F for the spring is shown in Fig. 2.4.

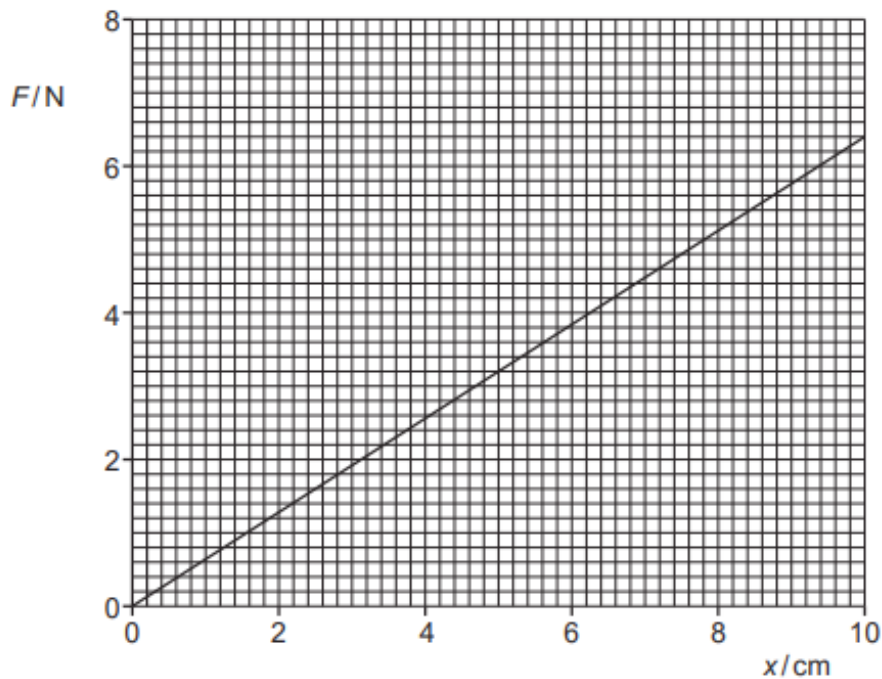


Fig. 2.4

The ball is a uniform sphere of steel of diameter 0.016 m and mass 0.017 kg.

- (i) Calculate the density of the steel.

density = kg m^{-3} [3]

- (ii) All of the elastic potential energy in the spring is converted into kinetic energy of the ball. The speed of the ball as it leaves the spring is 4.9 m s^{-1} .

Show that the maximum elastic potential energy of the spring is 0.20 J.

(iii) Use Fig. 2.4 to determine the spring constant k of the spring.

$$k = \dots\dots\dots \text{Nm}^{-1} \quad [2]$$

(iv) Use your answer in (b)(iii) and the value of energy given in (b)(ii) to determine the compression x_0 of the spring.

$$x_0 = \dots\dots\dots \text{m} \quad [2]$$

(c) The steel ball is replaced by a polystyrene ball of the same diameter but of much lower mass. The spring is given compression x_0 and is then released.

Air resistance on this ball is **not** negligible after it leaves the spring.

Explain:

(i) why this ball leaves the spring with a greater speed than that of the steel ball

.....
.....
..... [1]

(ii) why this ball takes a longer time to reach the ground than the steel ball.

.....
.....
..... [1]

[Total: 17]

40. 9702/22/O/N/22/Q4

A horizontal spring is fixed at one end. A block is pushed against the other end of the spring so that the spring is compressed, as shown in Fig. 4.1.

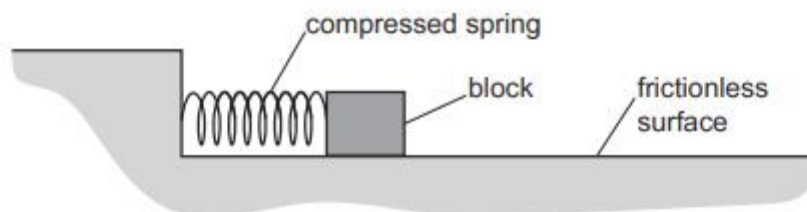


Fig. 4.1

The block is released and accelerates along a horizontal frictionless surface as the spring returns to its original length. The block leaves the end of the spring with a speed of 2.3 m s^{-1} , as shown in Fig. 4.2.

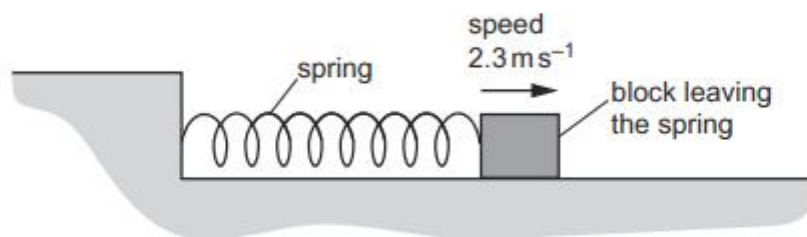


Fig. 4.2

The block has a mass of 250 g and the spring has a spring constant of 420 N m^{-1} .

Assume that the spring always obeys Hooke's law and that all the elastic potential energy of the spring is transferred to the kinetic energy of the block.

(a) Calculate the kinetic energy of the block as it leaves the spring.

kinetic energy = J [2]

(b) Calculate the compression of the spring immediately before the block is released.

compression = m [2]

- (c) After leaving the spring, the block moves along the surface until it hits a barrier at a speed of 2.3 m s^{-1} . The block then rebounds at a speed of 1.5 m s^{-1} and moves back along its original path. The block is in contact with the barrier for a time of 0.086 s .

Calculate:

- (i) the change in momentum of the block during the collision

change in momentum = N s [2]

- (ii) the average resultant force exerted on the block during the collision.

average resultant force = N [1]

- (d) The maximum compression x of the spring is now varied in order to vary the kinetic energy E_K of the block as it leaves the spring. Assume that all the elastic potential energy in the spring is always transferred to the kinetic energy of the block.

On Fig. 4.3, sketch a graph to show the variation with x of E_K .

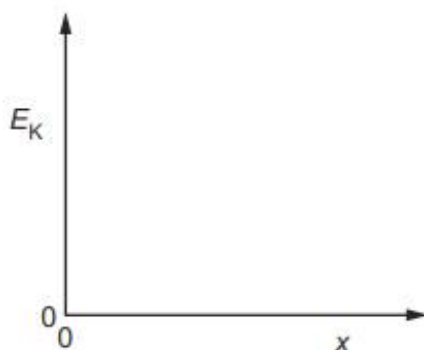


Fig. 4.3

[1]

[Total: 8]

41. 9702/23/O/N/22/Q3

(a) State the principle of moments.

.....

.....

..... [2]

(b) A hollow plastic sphere is attached at one end of a bar. The sphere is partially submerged in water and the bar is attached to a fixed vertical support by a pivot P, as shown in Fig. 3.1.

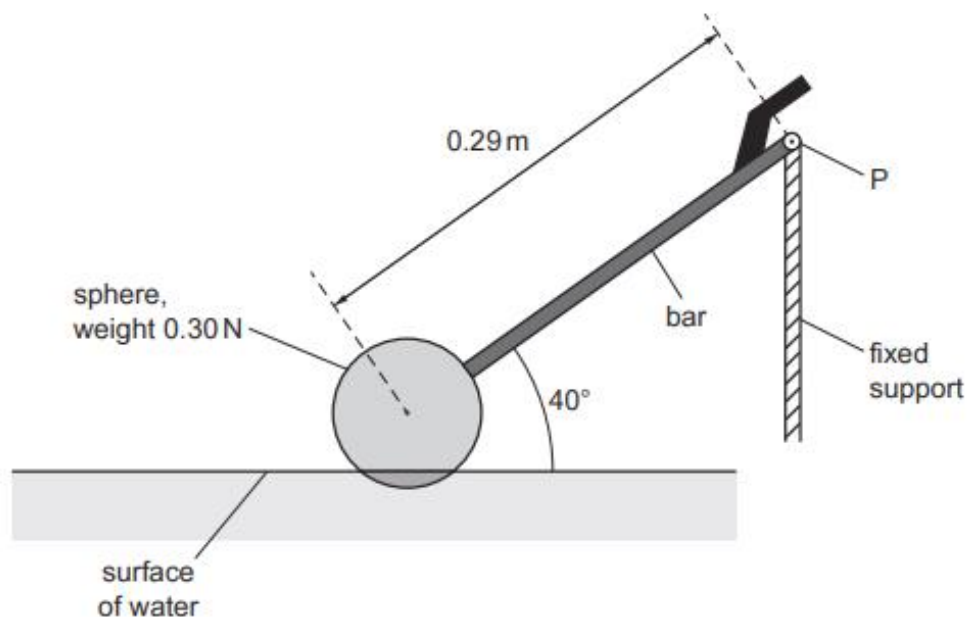


Fig. 3.1 (not to scale)

The sphere has weight 0.30 N. The distance from P to the centre of gravity of the sphere is 0.29 m. Assume that the weight of the bar is negligible.

Calculate the moment of the weight of the sphere about P.

moment = Nm [2]

- (c) The system shown in Fig. 3.1 is part of a mechanism that controls the amount of water in a tank.

Water enters the tank and causes the sphere to rise. This results in the bar becoming horizontal. Fig. 3.2 shows the system in its new position.

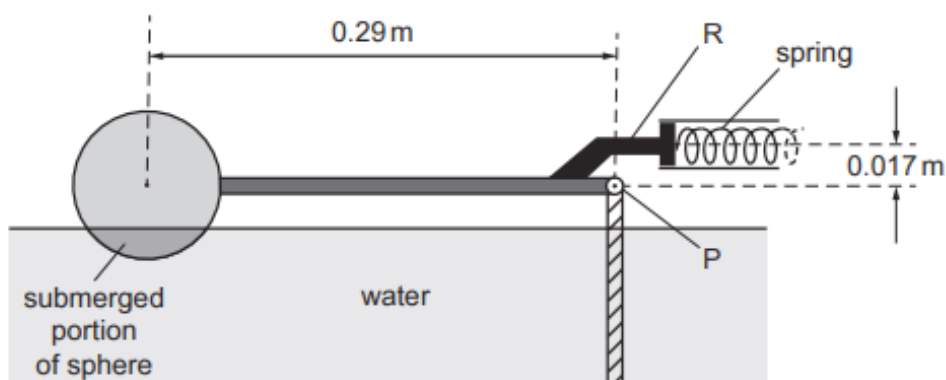
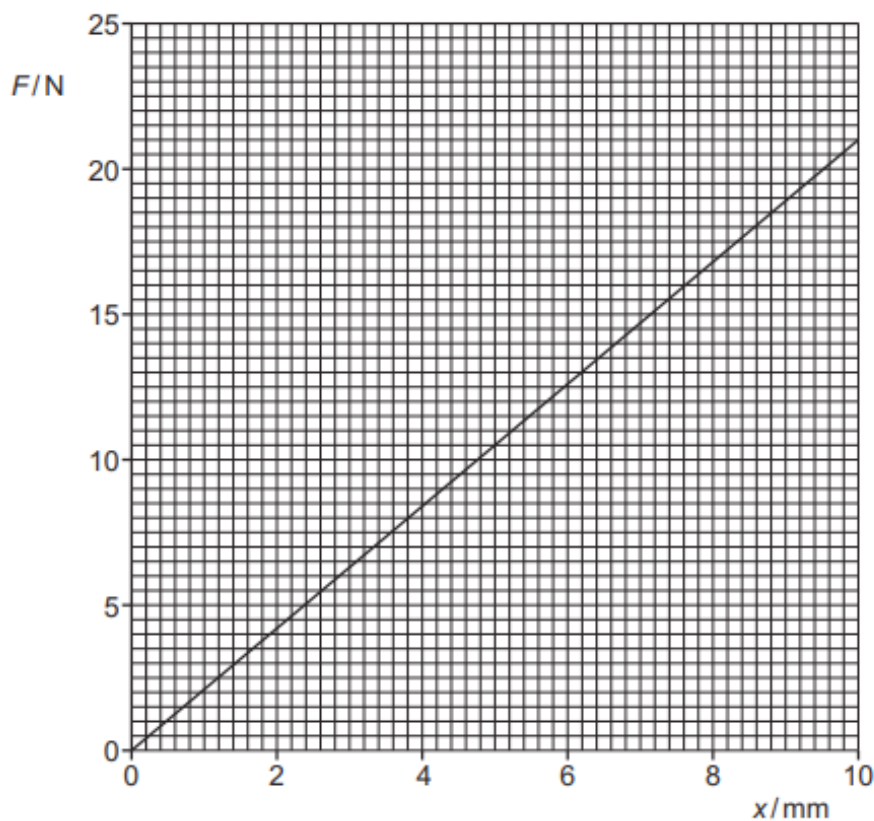


Fig. 3.2 (not to scale)

In this position the rod R exerts a force to compress a horizontal spring that controls the water supply to the tank. R is positioned at a perpendicular distance of 0.017 m above P.

The variation of the force F applied to the spring with compression x of the spring is shown in Fig. 3.3.



- (i) Use Fig. 3.3 to calculate the spring constant k of the spring.

$$k = \dots\dots\dots \text{Nm}^{-1} \quad [2]$$

- (ii) At the position shown in Fig. 3.2, the system is stationary and in equilibrium.

The radius of the sphere is 0.0480m and 26.0% of the volume of the sphere is submerged.

The density of water is $1.00 \times 10^3 \text{kgm}^{-3}$.

Show that the upthrust on the sphere is 1.18 N.

[2]

- (iii) By taking moments about P, determine the force exerted on the spring by the rod R.

$$\text{force} = \dots\dots\dots \text{N} \quad [2]$$

- (iv) Calculate the elastic potential energy E_p of the compressed spring.

$$E_p = \dots\dots\dots \text{ J [2]}$$

- (d) When the sphere moves from the position shown in Fig. 3.1 to the position shown in Fig. 3.2, the upthrust on the sphere does work.
Assume that resistive forces are negligible.

Explain why the work done by the upthrust is not equal to the gain in elastic potential energy of the spring.

.....
..... [1]

[Total: 13]

42. 9702/22/M/J/23/Q4

A spring is suspended from a fixed point at one end. The spring is extended by a vertical force applied to the other end. The variation of the applied force F with the length L of the spring is shown in Fig. 4.1.

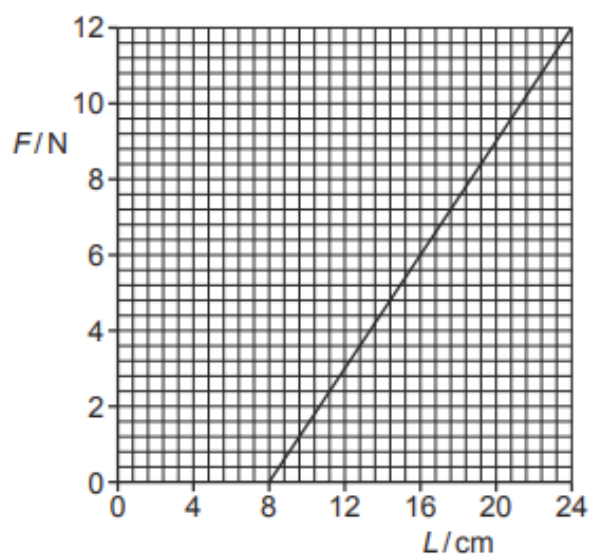


Fig. 4.1

For the spring:

- (a) state the name of the law that gives the relationship between the force and the extension

..... [1]

- (b) determine the spring constant, in Nm^{-1}

spring constant = Nm^{-1} [2]

(c) determine the elastic potential energy when $F = 6.0\text{ N}$.

elastic potential energy = J [2]

[Total: 5]

43. 9702/22/O/N/23/Q3

A vertical rod is fixed to the horizontal surface of a table, as shown in Fig. 3.1.

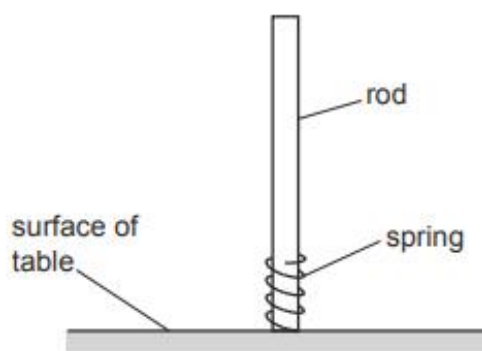


Fig. 3.1 (not to scale)

A spring of mass 7.5g is able to slide along the full length of the rod.

The spring is first pushed against the surface of the table so that it has an initial compression of 2.1 cm. The spring is then suddenly released so that it leaves the surface of the table with a kinetic energy of 0.048 J and then moves up the rod.

Assume that the spring obeys Hooke's law and that the initial elastic potential energy of the compressed spring is equal to the kinetic energy of the spring as it leaves the surface of the table. Air resistance is negligible.

- (a) By using the initial elastic potential energy of the compressed spring, calculate its spring constant.

spring constant = Nm^{-1} [2]

- (b) Calculate the speed of the spring as it leaves the surface of the table.

speed = ms^{-1} [2]

- (c) The spring rises to its maximum height up the rod from the surface of the table. This causes the gravitational potential energy of the spring to increase by 0.039 J.
- (i) Calculate, for this movement of the spring, the increase in height of the spring after leaving the surface of the table.

increase in height = m [2]

- (ii) Calculate the average frictional force exerted by the rod on the spring as it rises.

average frictional force = N [2]

- (d) The rod is replaced by another rod that exerts negligible frictional force on the moving spring. The initial compression x of the spring is now varied in order to vary the maximum increase in height Δh of the spring after leaving the surface of the table. Assume that the spring obeys Hooke's law for all compressions.

On Fig. 3.2, sketch a graph to show the variation with x of Δh . Numerical values are not required.

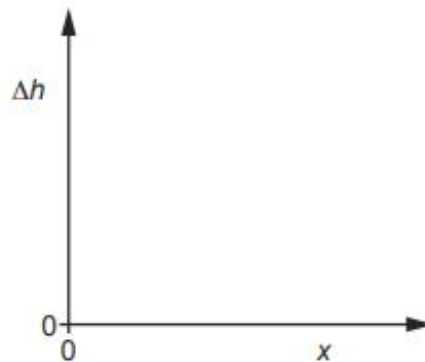


Fig. 3.2

[2]

[Total: 10]

44. 9702/23/O/N/23/Q4

Fig. 4.1 shows the variation with extension x of the tensile force F for two wires, G and H, made from the same material.

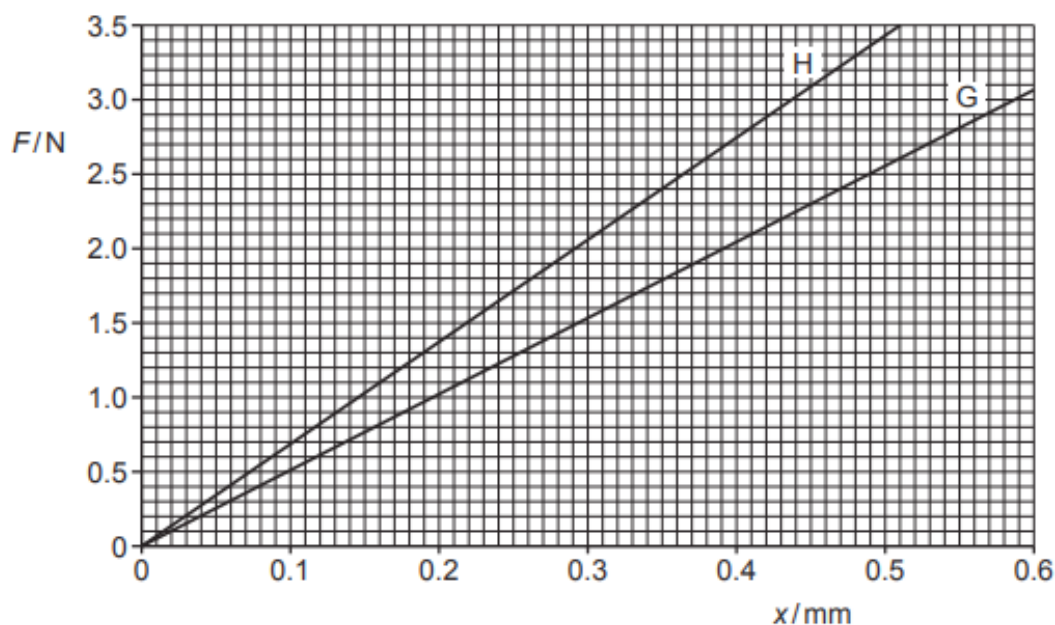


Fig. 4.1

The elastic limit has not been exceeded for G or H.

(a) For the lines in Fig. 4.1:

(i) state what is represented by the gradient

..... [1]

(ii) explain why the area under the line represents the elastic potential energy of the wire.

.....

 [2]

(b) Wires G and H are joined together end-to-end to form a composite wire of negligible weight. The composite wire hangs vertically from a fixed support.

A block of weight of 2.0 N is attached to the end of the wire, as shown in Fig. 4.2.

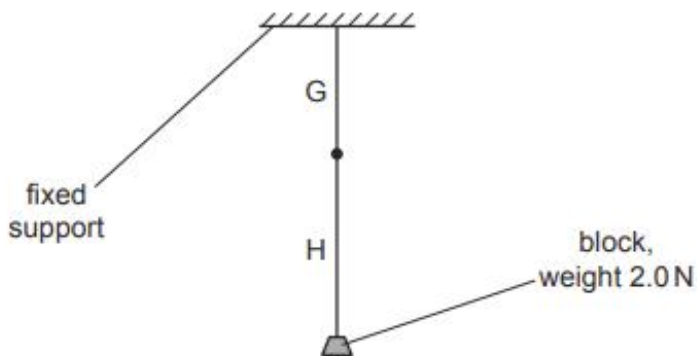


Fig. 4.2

(i) Use Fig. 4.1 to determine:

- the extension x_G of wire G

$$x_G = \dots\dots\dots \text{ mm}$$

- the extension x_H of wire H.

$$x_H = \dots\dots\dots \text{ mm}$$

[1]

(ii) Calculate the total elastic potential energy E_p of the composite wire due to the weight of the block.

$$E_p = \dots\dots\dots \text{ J} \quad [2]$$

(iii) The original length of wire G is L and the original length of wire H is $1.5L$.

Calculate the ratio

$$\frac{\text{cross-sectional area of wire G}}{\text{cross-sectional area of wire H}}.$$

$$\text{ratio} = \dots\dots\dots [3]$$